Scaling Science? Research Capacity and Scientific Production within a Healthcare Organization*

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Abstract

This study focuses on how an expansion in organizational research capacity affects scientific production within a large Italian university hospital. In 2018, the hospital was recognized as a Scientific Institute for Research, Hospitalization and Healthcare (IRCCS), gaining access to additional public funding, research infrastructure through formal research lines, and organizational support. Using physician-level panel data for 2012–2022 and difference-in-differences designs, we find a large and persistent increase in publications among physicians exposed to the new research environment. The increase is concentrated among physicians who were already more research-active before recognition and occurs mainly within established scientific fields and existing collaboration networks. Field shares, the relative importance of new collaborations, and authorship patterns change little. Total citations increase due to the shift in volume, while we find no robust evidence of a change in citations per publication. Complementary and descriptive evidence from hospital-level clinical indicators suggests an expansion in high-complexity procedure volumes following recognition, but no systematic variations in clinical outcomes. Therefore, we show that expanding research resources can generate substantially more science without fundamentally changing how science is produced.

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1 Introduction

Evaluations of public support for science typically focus on its effects on publications, citations, or patents. However, increases in scientific output may reflect different organizational

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processes: additional resources may allow established researchers and teams to expand existing activities or may also foster new collaborations, capabilities, and research directions. We distinguish between *scaling*, whereby scientific production expands along existing trajectories, and *transformation*, whereby additional capacity changes who conducts research, with whom, and in which scientific domains. This distinction is analytical rather than normative: transformation is not necessarily an objective of research funding, nor is it inherently preferable to scaling. Its relevance depends on whether policymakers seek only to expand scientific output or also to alter its composition and organization. Distinguishing between the two responses nevertheless allows us to identify the organizational margins through which additional research capacity generates scientific output.

The distinction is particularly relevant in healthcare, where scientific knowledge advances through diffusion across researchers and clinical units (Barrenho et al., 2021; Chandra & Staiger, 2007), collaborative networks (Barrenho et al., 2025), the attraction of skilled workers (Maynou et al., 2024), and institutional changes (Ghandour et al., 2022). As a matter of fact, biomedical research largely depends on how personnel and organizational inputs are combined, on top of the mere financial resources. For instance, existing evidence highlights the importance of research-team composition for lab productivity (Carayol & Matt, 2004), and the complementarity of expertise, stable funding, and external resources for scientific creativity (Heinze et al., 2009).

Developing new medical research capabilities also requires substantial investment, often involving high short-run costs relative to observable benefits (Grant & Buxton, 2018). Moreover, biomedical research relies on capital-intensive technologies and infrastructure, whose long-run returns may depend on the technologies developed and their diffusion among practitioners and patients (Chandra & Skinner, 2012). These characteristics suggest that additional research capacity may be absorbed unevenly, with researchers and teams possessing greater prior knowledge and established capabilities better positioned to exploit new resources (Cohen & Levinthal, 1990). Consequently, increased resources may reinforce existing research trajectories and specializations mainly, rather than re-direct or transform the scientific activity, thus causing substantial output increases without changes in the organization of research.

We investigate these questions in a large Italian university hospital that obtained IRCCS status (*Istituto di Ricovero e Cura a Carattere Scientifico*) in February 2018, a formal recognition granted by the Italian Ministry of Health to hospitals combining high-quality clinical care with specialized biomedical research. The recognition granted access to recurring public funding for research purposes, organized around formally approved research lines (broadly defined to accommodate research across different domains), direct eligibility for national competitive project-based research programs alongside Italian regions, and additional infrastructure and organizational support. Most of the additional funding is allocated di-

rectly to the hospital, which retains discretion over its distribution across research activities and within the broad research lines, whereas competitive project-based funding constitutes a separate channel. Unlike individual grant programs, the institutional funding scheme for recurring funding does not assign fixed amounts to individual physicians, which means that the treatment exposure is therefore based on a group-level distinction. We link hospital personnel records to Web of Science data on publications, citations, authorship positions, coauthors, and scientific fields, constructing a balanced physician-level panel for 2012–2022.

Our baseline difference-in-differences design compares changes in scientific output between physicians included in the institutional IRCCS research perimeter and those outside of it. We classify physicians ever included in the perimeter as exposed from recognition onward. Therefore, we do not interpret the estimates as returns to individual funding transfers, as they are indeed intention-to-treat (ITT) effects of access to expanded organizational research capacity. Event-study estimates show no systematic differential pre-trends and a gradual, growing response after the recognition, clearly consistent with publication lags. Annual publication output of the physicians exposed to the IRCCS environment increases substantially, by approximately 70% relative to the treated pre-period mean, with effects persisting through the end of the sample.

The composition of this output growth is the paper's main contribution. First, the response is strongly biased by existing capabilities. Physicians who were already publishing and those in higher pre-treatment productivity groups experience the largest gains, while the effect on the extensive margin (entry into publishing) is limited if not totally absent. Second, output rises across several established medical fields, but both the organization's and the researchers' field portfolios remain stable. Third, internal and external collaboration expand through greater activity within existing networks. On the other side new links increase in number, but they do not gain weight relative to existing relationships. Fourth, last-authored and even more middle-authored publications increase more clearly than first-author publications, providing little evidence of a relevant shift toward new scientific leadership. Finally, total (discounted) citations rise with publication volume, while citations per publication do not display relevant shifts. Our findings are informative to the extent that they show how recognition primarily scaled up the existing research activities rather than reallocating its knowledge base. Complementary, additional evidence from hospital-level clinical indicators also points to an expansion in high-complexity procedure volumes, but no systematic improvement in clinical outcomes; given the limitations of the hospital-level comparisons though, the latter results remain suggestive without allowing for robust causal interpretation.

The paper contributes to three related strands of research. First, a large literature examines the effects of public research funding on scientific production, innovation, and researchers' careers. Several studies find positive effects on publication output and scientific impact

(Azoulay et al., 2011; Benavente et al., 2012), although their magnitude and robustness vary across funding schemes and empirical designs (Ghirelli et al., 2023; Jacob & Lefgren, 2011). At the institutional level, Banal-Estañol et al. (2023) find that performance-based funding increases aggregate research output without improving productivity per researcher. Beyond publication counts, Babina et al. (2023) show that reductions in public research funding decrease publications while shifting innovation toward patenting and private appropriation. Funding can also affect researcher mobility and the allocation of scientific talent (Baruffaldi et al., 2020), while Robbiano (2022) examines how establishing a publicly funded research institute affects regional innovation and human capital. We contribute to this literature by estimating the effects of an institution-wide expansion in publicly funded research capacity on physicians' scientific productivity and citation impact, instead of examining the returns to individual research grants.

Second, our study relates to the resources and organizational conditions shaping scientific production. Previous research highlights the importance of research-team composition and organizational environments for scientific productivity and creativity (Carayol & Matt, 2004; Heinze et al., 2009). More recently, Baruffaldi and Gaessler (2026) show that specialized physical capital generates persistent returns to scientific production while potentially reinforcing existing research trajectories. These factors are particularly relevant in healthcare, where research and clinical activities rely on complementary inputs and scientific production spans academic and clinical boundaries (Barrenho et al., 2025). Related studies examine how research investments attract skilled workers and shape healthcare organizations (Ghandour et al., 2022; Maynou et al., 2024). Closest to our setting, Azoulay et al. (2025) show that reductions in clinical reimbursements increased research activity across US academic medical centers, alongside changes in the composition of scientific production. We complement this evidence by examining how an increase in funding directly allocated to research affects scientific production within a single hospital, tracing the response through collaboration networks and authorship structures *within* the institution, rather than *across* organizations.

Third, we contribute to the literature on how research resources and incentives shape the allocation of scientific activity across researchers, tasks, and research directions. Funding arrangements can influence scientific exploration and novelty (Azoulay et al., 2011; Wang et al., 2018), while researchers' characteristics may shape the questions investigated (Dossi, 2024) and the benefits of research inputs. For instance, Lepinteur and Nisticò (2025) find that access to administrative data improves research visibility and career prospects without increasing publication output, with heterogeneous benefits across researchers. Research and career incentives can also shift effort across competing tasks, consistent with the multitasking framework of Holmstrom and Milgrom (1991). In academia, De Philippis (2021) document a shift from teaching to research following stronger research incentives, while Checchi et

al. (2021) find heterogeneous responses to promotion incentives depending on researchers' abilities. Such trade-offs are particularly relevant in healthcare, where physicians combine clinical care and research. As previously discussed, Azoulay et al. (2025) find that reductions in clinical reimbursements increase research activity in US academic medical centers, suggesting a reallocation of resources and effort between clinical and scientific production. Our research examines whether additional research capacity alters the distribution of scientific activity across physicians with different initial capabilities and redirects production toward new research fields, or mainly reinforces existing specialization patterns, while also exploring whether the expansion in scientific output is accompanied by changes in clinical activity and patient outcomes.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting, and Section 3 presents the data. Section 4 outlines the identification strategy. Section 5 reports the main effects of IRCCS recognition on scientific output, including the analysis of staggered entry into the research perimeter. Section 6 presents the identification validity and robustness checks, while Section 7 examines heterogeneity by physicians' pre-treatment research productivity. Section 8 investigates changes in research specialization, collaboration networks, authorship positions, and scientific impact. Section 9 concludes.

2 Institutional Framework

The study is conducted at a large private university hospital in Rome that has achieved high international standards in specialized healthcare, medical education, and biomedical research in recent years. In February 2018, the Italian Ministry of Health recognized the hospital as an *Istituto di Ricovero e Cura a Carattere Scientifico* (IRCCS, i.e. Scientific Institute for Research, Hospitalization and Care) in the broad subjects of *Personalized Medicine* and *Innovative Biotechnologies*. In Italy, IRCCSs are a specific class of research-intensive hospitals within the National Health Service. They combine highly specialized clinical care with biomedical, clinical, and predominantly translational research (Ministry of Health, 2026). In the present context, translational research denotes the duality between scientific research and clinical assistance: clinical needs shape research trajectories, while scientific discoveries and innovations shall be directly translated into patient care. The IRCCS model supports this connection by locating clinical activity, research personnel, and scientific infrastructure within the same organization.

Recognition is granted by the Ministry of Health conditional on scientific, clinical, technological, and organizational requirements and is subject to periodic reassessment (D.lgs 200, 2022; Ministry of Health, 2025b). It is neither an automatic consequence of teaching-hospital status nor a purely administrative designation. Applicants must document excellence in

clinical care and scientific production, together with appropriate research personnel, infrastructure, equipment, funded projects, clinical trials, biobanks, collaborations, and dedicated research accounts. Appendix Table A1 summarizes the principal requirements. Although the IRCCS framework is specific to Italy, related institutional arrangements exist in other countries. Close examples include the French *Instituts Hospitalo-Universitaires* and the Spanish *Institutos de Investigación Sanitaria*, which integrate hospital care, biomedical research, and dedicated research governance. German university hospitals and US academic medical centers perform broadly comparable functions, although they are not based on an equivalent national designation.

Procedurally, IRCCS recognition follows a formal evaluation involving regional and national authorities. Applications are submitted to the relevant region, which verifies their consistency with regional health planning. Once regionally approved, applications are assessed by a ministerial expert commission, which may conduct site visits before the final decision. In the studied case, the Lazio Region transmitted the application in October 2017, the evaluation commission conducted site visits in December 2017 and January 2018, and the hospital was formally recognized on February 28, 2018 in the two complementary areas of Personalized Medicine and Innovative Biotechnologies. The interval between formal submission and recognition was therefore relatively short, limiting the period in which physicians could adjust their publication activity in anticipation of the final decision. Although formally granted in two specific disciplines, the hospital implemented them through broad research lines spanning multiple clinical and scientific specialties, allowing for great flexibility in the range of IRCCS-related research. The IRCCS status is subsequently subject to periodic reassessment by experts of the underlying requirements. The main steps of the recognition process are summarized in Appendix Table A2.

The recognition changed the hospital's research environment in several ways. First and foremost, in terms of financial resources. The IRCCS status granted access to recurring *Current Research* funding, a dedicated annual stream provided by the Ministry of Health and allocated to IRCCSs on the basis of their scientific and clinical activity. For the studied hospital, this funding amounted to roughly €6–7 million per year in the early 2020s, reaching €7.4 million in 2022. IRCCS research activities are organized through institution-specific research lines approved within the Ministry's three-year programming framework (Ministry of Health, 2025a)[1]. The funding can also be employed for research equipment and technological infrastructure. A second funding-related advantage obtained through IRCCS recognition is the possibility to participate directly in national competitive research programs, including

[1]As mentioned already, the formal research lines were defined broadly enough to accommodate a very heterogeneous set of activities in practice, ranging from omics, predictive modelling, big-data applications to ageing, surgery, intensive care, medical devices, nanomedicine, and women's and child health, among others.

Targeted Research, which finances project-based research across the National Health Service[2]. The additional resources supported a formal organization of research activity, including dedicated scientific management, accounting and reporting procedures, administrative support, and an official *research perimeter*. This organizational dimension has been tackled by previous work already, which distinguished project-oriented from organization-oriented research engagement within IRCCS hospitals (Mollica et al., 2022).

The *research perimeter* is designed by the hospital management, and formally identifies physicians participating in the IRCCS environment. It consists mainly, but not exclusively, of academically affiliated physicians (i.e., physicians employed as faculty members in the university affiliated to the hospital), and is updated administratively over time. Our baseline definition classifies physicians ever included in the perimeter as exposed from the 2018 recognition. We additionally exploit the timing of formal entry into the perimeter in a complementary staggered-entry analysis and assess robustness to alternative treatment-timing definitions[3].

3 Data

We construct a panel dataset of the professionals working in the hospitals, combined with their research output spanning 2012 to 2022. The core personnel data come from the hospital's administrative records, which include all individuals structurally employed in a professional capacity (physicians, researchers, and other healthcare staff management) from 2017 onward. From that, we identify our population of interest: physicians and medical researchers continuously employed at the hospital between 2017 and 2022. For each person, the HR data provide demographic information (age, gender), employment details (job title or role, hiring date), and indicate whether the individual has an academic affiliation (university-employed physician) or is a hospital-only employed clinician. Data on medical interns, residents, PhD students, and external collaborators were not available, and they are therefore excluded by construction.

Because the available employer records begin in 2017, we supplemented them with hire dates and career information obtained from institutional records, CVs, professional registries,

[2]Under the ordinary framework, Targeted Research funds are generally accessed through institutional recipients such as the regions and the National Institute of Health (ISS), which then support projects within their respective networks. IRCCSs, by contrast, can participate directly in these competitive calls alongside the Regions, the ISS, and the Experimental Zooprophylactic Institutes (IIZZSS).

[3]A publication-based Management-by-Objectives (MBO) bonus was also introduced in 2017 for non-academic physicians (i.e., doctors with no structural and permanent affiliation to the university), to foster their research activity and their integration to the forthcoming IRCCS environment. Given that the eligibility depended on employment status rather than IRCCS participation, we treat the scheme only as institutional background and as a robustness dimension, not as part of the main treatment definition.

and other publicly available sources. This information allows us to retrospectively reconstruct physicians' affiliation histories and extend the panel back to 2012. Since workers who left the hospital before 2017 cannot be identified, the baseline analysis relies on a balanced panel of individuals continuously observed throughout the study period, avoiding attrition concentrated in the post-treatment years[4]. The resulting balanced panel covers 584 physicians and researchers observed annually from 2012 to 2022, yielding 6,424 person-year observations[5]. Given that some physicians entered the IRCCS perimeter after 2018, the baseline analysis adopts an ever-treated definition; alternative specifications account explicitly for staggered formal entry.

	IRCCS			Non-IRCCS			Total		
	Mean	SD	N	Mean	SD	N	Mean	SD	N
Annual publications	4.32	(6.27)	2,739	0.64	(1.40)	3,685	2.21	(4.60)	6,424
Citations (Web of Science)	122.40	(288.69)	2,739	11.14	(33.91)	3,685	58.58	(198.03)	6,424
Citations (all outlets)	129.96	(306.13)	2,739	11.85	(35.92)	3,685	62.21	(210.00)	6,424
Age	52.06	(8.00)	2,739	50.43	(7.68)	3,685	51.13	(7.86)	6,424
Female	0.37	(0.48)	2,739	0.46	(0.50)	3,685	0.42	(0.49)	6,424
Tenure	17.28	(10.33)	2,739	15.55	(9.71)	3,685	16.29	(10.01)	6,424
Co-authors per published paper	10.66	(6.83)	2,183	8.74	(6.95)	1,202	9.98	(6.93)	3,385

Table 1: Physician-year descriptives by IRCCS exposure

To measure research output, we gathered data on scientific publications from Clarivate's Web of Science database[6]. We retrieved all publications (articles, reviews, etc.) from 2012 to 2022 for the authors in both the balanced and unbalanced panels[7]. Each publication record provides the publication year, journal, number and name of co-authors, citations (as of the time of data collection, October 2024), and other bibliometric information. From this matched dataset, we construct our main outcome variable, the number of publications per person per year. Publications co-authored by multiple physicians in our sample are counted once for each contributing physician when constructing individual-year outcomes, as is standard for author-level publication measures. By contrast, whenever we report institution-level

[4]The hire dates are missing for individuals who were not employed in 2020 and/or 2023. Additional details for such missing units were collected from institutional websites, CVs, professional registries (FNOMCeO), LinkedIn, and other publicly available sources. Details on the procedure are available upon request.

[5]As robustness check, we construct an unbalanced panel incorporating physicians who were not continuously employed between 2012 and 2022. Such panel consists of 1,540 units. Appendix Figure A2 reports the annual composition of the unbalanced sample and of the IRCCS perimeter.

[6]We use Web of Science because it is the bibliometric database employed by the hospital to evaluate its scientific performance and assess research-related performance incentives.

[7]We carefully ensured that the recovered authors were affiliated to the hospital, by matching names and surnames, fields of research, and double checking all of them, to avoid homonyms. This reduces the risk of homonym-related matching errors relative to matching based on researchers' surnames alone (Jacob & Lefgren, 2011).

statistics, each publication is counted only once regardless of the number of co-authors.

We also record the total citations received by each individual's publications in order to observe research impact. In addition, the bibliometric information allows us to construct several measures of research collaboration, including the number of co-authors per publication, the number of internal collaborators affiliated with the hospital, and the entry of new collaborators into researchers' co-authorship networks. These variables are used later in the paper to explore potential mechanisms behind the estimated effects and to perform robustness checks.

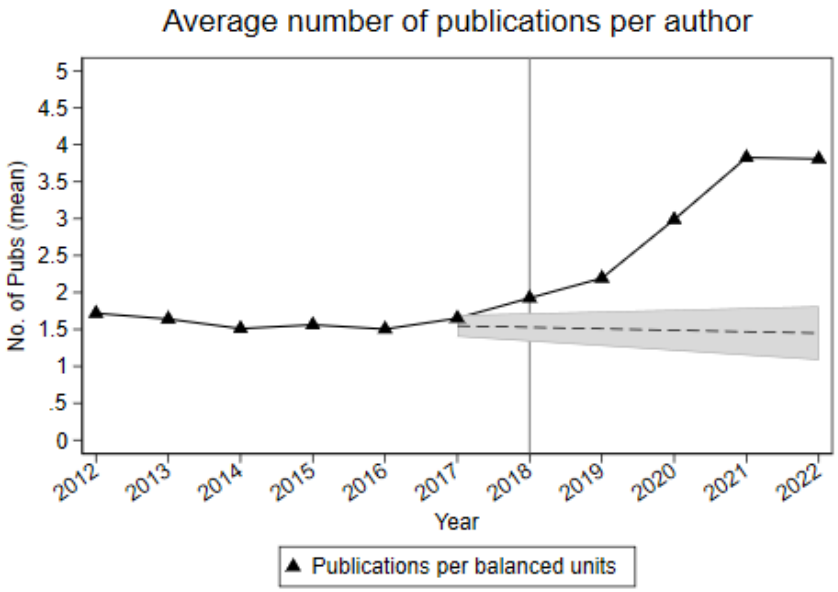


Figure 1: Average publications per researcher per year (2012–2022), balanced panel. Publications co-authored by multiple sampled physicians contribute to each author's individual publication count. The 2018 vertical line marks IRCCS recognition. The dashed line extrapolates the pre-recognition linear trend and the shaded area reports its 95% confidence interval.

Table 1 reports summary statistics for the balanced sample. The relevant descriptive distinction is whether physicians are ever included in the IRCCS research perimeter. IRCCS-exposed physicians display markedly higher research activity, averaging 4.32 publications per year compared with 0.64 among non-IRCCS physicians, and they also accumulate more citations. They are slightly older, have a slightly longer experience within the institution, and are less likely to be female; they also collaborate with a larger number of co-authors per paper. These level differences confirm that IRCCS exposure is concentrated among physicians who were already more research-oriented, underscoring the importance of relying on changes over time rather than cross-sectional comparisons. The timing of formal entry is examined separately in the staggered-entry analysis.

Figure 1 plots the average number of publications per physician per year between 2012

and 2022 for the balanced panel. Publication activity remains stable until 2017, increasing markedly after 2018. Appendix Figure A1 shows that the same expansion is visible in the annual number of unique publications produced by researchers in the balanced sample, while Appendix Figure A3 shows that the average pattern is robust to various degrees of attrition.

4 Identification Strategy

Baseline

We estimate the effect of IRCCS recognition on scientific production using a difference-in-differences design comparing physicians ever included in the IRCCS research perimeter with physicians never included, before and after the hospital's recognition in 2018. Our baseline specification, estimated via OLS, is:

$$Y_{it} = \alpha + \theta \left(Post2018_t \times IRCCS_i \right) + \delta_i + \lambda_{d(i,t),t} + \varepsilon_{it}, \tag{1}$$

where Y_{it} denotes the scientific output of physician i in year t, $Post2018_t$ equals one from 2018 onward, and $IRCCS_i$ identifies physicians ever included in the IRCCS research perimeter. Physician fixed effects, δ_i, absorb time-invariant individual heterogeneity, while department-by-year fixed effects, $\lambda_{d(i,t),t}$, absorb shocks shared by physicians working in the same department in a given year. Standard errors are clustered at the physician level. Given that the individual receipt of funding is not observed, we classify physicians ever included in the research perimeter as exposed from the date of institutional recognition and interpret θ as the intention-to-treat effect of exposure to the IRCCS research environment. The coefficient captures the combined institutional change, and not the return to a specific funding component. We consider two outcomes to identify scientific output: annual publication counts and an indicator equal to one if a physician publishes at least once in a given year and 0 otherwise. The latter allows us to assess whether the expansion in scientific production also operates through the extensive margin.

The baseline analysis uses a balanced panel covering 2012–2022. Following the same physicians throughout the observation period prevents changes in the composition of the workforce from mechanically generating the estimated post-recognition differential. However, we also assess the sensitivity of the results to this restriction by using the full unbalanced panel. Identification also requires that, in the absence of IRCCS recognition, publication output among IRCCS and non-IRCCS physicians would have evolved in parallel. The groups may however differ in their initial publication levels; what matters is that they do not exhibit systematically different trends before recognition. Our fixed-effects specification therefore identifies the effect from changes in publication output within physicians over time.

To examine pre-treatment dynamics and the evolution of the effect after recognition, we estimate the following event-study specification:

$$Y_{it} = \alpha + \sum_{s \neq 2017} \beta_s \left[IRCCS_i \times \mathbb{1}(t = s) \right] + \delta_i + \lambda_{d(i,t),t} + \varepsilon_{it}, \tag{2}$$

where 2017 is the omitted reference year. The coefficients for 2012–2016 capture differences in pre-recognition trends relative to 2017, while those for 2018–2022 trace the timing and persistence of the response following IRCCS recognition.

Staggered Entry

The baseline specification classifies physicians ever included in the IRCCS perimeter as exposed from institutional recognition in 2018 onward. As an alternative treatment definition, we use each physician's first recorded year of formal entry into the perimeter. We begin by applying the decomposition proposed by Goodman-Bacon (2021) to a conventional static TWFE specification estimated on the balanced sample. The decomposition expresses the estimated coefficient as a weighted average of all available two-group, two-period difference-in-differences comparisons. It thus reveals how much of the coefficient is generated by comparisons between treated and never-treated physicians and how much is generated by comparisons among cohorts entering the IRCCS perimeter at different times. We use this exercise only to characterize the comparisons underlying the full-sample TWFE coefficient; we do not report it as a separate static treatment-effect estimate.

The main staggered analysis instead focuses on treatment dynamics and restricts the sample to physicians who eventually enter the perimeter. The identification in this specification therefore relies on differences in formal entry timing. We estimate the following TWFE event-study specification:

$$Y_{it} = \alpha + \sum_{k \neq -1} \beta_k \mathbb{1}(t - G_i = k) + \delta_i + \lambda_{d(i)t} + \varepsilon_{it}, \tag{3}$$

where G_i is physician i's first recorded year of entry and $k = t - G_i$ denotes event time. The year preceding formal entry, $k = -1$, is the omitted reference period. The coefficients β_k trace publication dynamics before and after entry relative to that year, conditional on physician fixed effects δ_i and department-by-year fixed effects $\lambda_{d(i)t}$, which absorb time-varying shocks common to physicians within the same department.

Given that conventional TWFE event-study estimates may be biased under heterogeneous timing and treatment effects, we also implement the estimators proposed by Callaway and Sant'Anna (2021), Gardner (2022), and Sun and Abraham (2021)[8]. All specifications include

[8]The three estimators address heterogeneity through different approaches. Callaway and Sant'Anna esti-

physician and department-by-year fixed effects, except for Callaway and Sant'Anna, which is estimated without department-by-year fixed effects.[9]. Standard errors are clustered at the physician level.

5 Results

5.1 Baseline Effects

Table 2 reports the estimated effect of IRCCS recognition on annual publication output across alternative sample definitions. Column (1) presents our preferred specification using the balanced physician panel, physician fixed effects, and department-by-year fixed effects. Columns (2)–(4) examine the sensitivity of the estimates to the composition of the physician sample. Specifically, Column (2) excludes physicians in the top 1% of the distribution of average annual publication output measured over the pre-treatment period (2012–2017). Column (3) restricts the sample to physicians with at least one pre-treatment publication, while Column (4) jointly imposes both restrictions[10]. Finally, Column (5) uses the full unbalanced panel to assess the sensitivity of the results to the balanced-panel requirement.

The baseline estimate in Column (1) indicates that IRCCS exposure increases annual publication output by 2.186 publications per physician-year relative to physicians outside the research perimeter. Compared with a pre-treatment mean of 3.12 publications among exposed physicians, this corresponds to an increase of approximately 70%, indicating a substantial expansion in scientific production following institutional recognition.

The sensitivity analyses give insight about the distribution of the response. Excluding the most prolific physicians, identified exclusively on the basis of their pre-treatment research output, reduces the estimated coefficient from 2.186 to 1.851 publications per physician-year, a decline of approximately 15%. This suggests that physicians with high initial research productivity contribute disproportionately to the magnitude of the aggregate response. Nonetheless, the effect remains positive, statistically significant, and comparatively substantial in relative terms after their exclusion; as a matter of fact, even by trimming the top 1%, the estimated

mate group-time average treatment effects using given comparison groups, aggregating them by event time. Gardner's two-stage estimator first estimates untreated potential outcomes using observations from units not-yet-treated, and then estimates dynamic effects from residualized outcomes. Sun and Abraham estimate cohort-specific event-time effects and subsequently aggregate them, avoiding the contamination of TWFE event-study coefficients. All three estimators are here implemented on the sample of ever-treated physicians. Callaway and Sant'Anna use not-yet-treated physicians as control groups, Gardner's first stage relies on not-yet-treated units, whereas Sun and Abraham use the last-treated cohort as the control group.

[9]Our implementation of the Callaway and Sant'Anna estimator does not directly accommodate high-dimensional department-by-year fixed effects. Its estimates should therefore be interpreted as complementary evidence rather than directly comparable estimates under an identical fixed-effects structure.

[10]Descriptive statistics for the restricted sample excluding the top 1% of pre-treatment publishers and physicians with no pre-treatment publications are reported in Appendix Table A3.

effect amounts to approximately 70% of the pre-treatment average output for the treated in the restricted sample. Restricting the sample to physicians with at least one pre-treatment publication yields an estimate of 2.194, while jointly imposing both restrictions produces an effect of 1.845 publications per physician-year. Therefore, even though the estimated coefficients indicate that the magnitude of the effect is partly influenced by top-tier researchers, the positive response extends beyond this small group, but preserves comparable sizes in relative terms across heterogeneous samples. At the same time, the table shows how the impact is not even driven by physicians entering scientific publishing after recognition. Column (5) reports eventually an estimated increase of 1.91 publications per physician-year, equivalent to approximately 62% of the pre-treatment mean in the dataset with attrition, suggesting that the main findings are not particularly sensitive to the requirement that physicians be continuously observed in the available timespan.

	(1) Baseline	(2) Trim 1%	(3) Pre-treatment publishers	(4) Trim 1% and publishers	(5) Unbalanced panel
Panel A: Annual publication count					
Post-2018 × IRCCS	2.186***	1.851***	2.194***	1.845***	1.944***
	(0.226)	(0.164)	(0.248)	(0.188)	(0.172)
Pre-treatment mean	3.12	2.63	3.38	2.85	3.07
Panel B: Any publication					
Post-2018 × IRCCS	0.012	0.014	0.032	0.035	0.022
	(0.022)	(0.022)	(0.026)	(0.027)	(0.019)
Pre-treatment mean	0.75	0.74	0.81	0.80	0.74
Observations	6,424	6,347	4,675	4,598	10,902
Physician FE	✓	✓	✓	✓	✓
Department × Year FE	✓	✓	✓	✓	✓

Table 2: IRCCS recognition and annual publication output. The table reports difference-in-differences estimates of the effect of IRCCS recognition in 2018, comparing physicians ever included in the IRCCS research perimeter with those never included. Panel A uses the annual number of publications per physician as the dependent variable; Panel B uses an indicator equal to one if the physician publishes at least once in a given year. Column (1) uses the balanced panel (2012–2022). Column (2) excludes physicians in the top 1% of mean annual publication output during 2012–2017. Column (3) retains physicians with at least one publication before 2018, while Column (4) combines both restrictions. Column (5) uses the full unbalanced panel. Pre-treatment means refer to exposed physicians in each estimation sample. All specifications include physician and department-by-year fixed effects. Standard errors, clustered at the physician level, are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Turning to the extensive margin, Panel B shows no statistically significant effects on the probability of publishing at least once in a given year. The estimated coefficients range from 0.8 to 3.3 percentage points across specifications and remain statistically indistinguishable from zero, despite relatively high pre-treatment publication probabilities among exposed

physicians (74–81%).

5.2 Dynamic Effects

Figure 2 reports the event-study estimates from Equation (2). Alongside the baseline balanced-panel specification, we consider the same three alternative sample definitions: excluding physicians in the top 1% of average annual pre-treatment publication output, restricting the sample to physicians with at least one pre-treatment publication, and jointly imposing both restrictions.

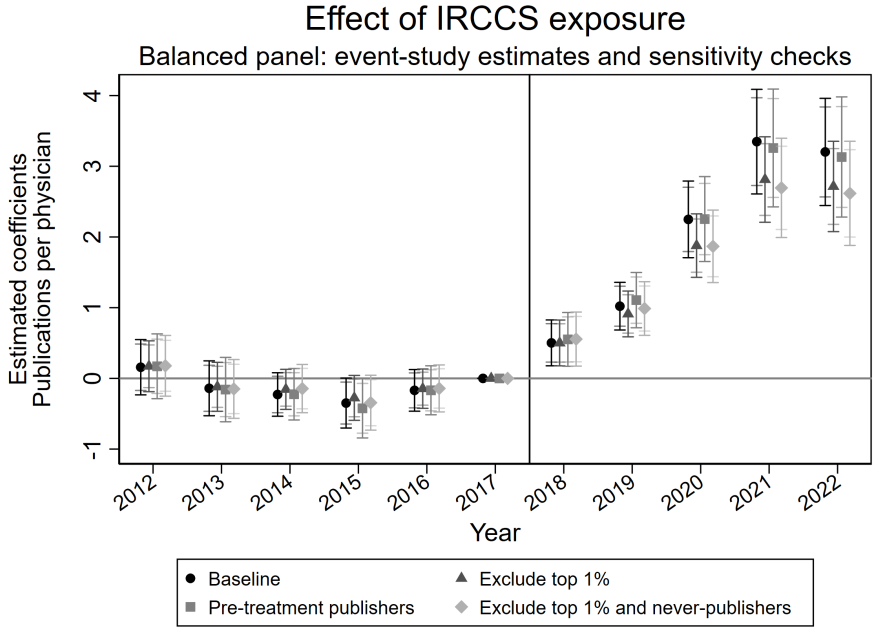


Figure 2: IRCCS recognition and scientific production: dynamic effects. The figure reports event-study estimates of the effect of IRCCS recognition on annual publication counts, comparing physicians ever included in the IRCCS research perimeter with those never included. Estimates are based on the balanced physician panel (2012–2022). The baseline specification is compared with three sample restrictions: excluding the top 1% of physicians by average annual pre-treatment publication output (2012–2017), retaining only physicians with at least one pre-treatment publication, and combining both restrictions. All specifications include physician and department-by-year fixed effects. Standard errors are clustered at the physician level. The reference year is 2017, with its coefficient normalized to zero. Vertical bars report 90% and 95% confidence intervals. The vertical line marks IRCCS recognition in 2018.

The estimates provide little evidence of systematic differential trends in publication output between IRCCS and non-IRCCS physicians before recognition. Although some differences are observed in the earlier pre-treatment years, the coefficients remain relatively stable and close to zero in the years immediately preceding 2018, across the alternative sample definitions. We further assess the robustness of the parallel trends assumption in Section 6.

Following recognition, publication output increases progressively. The estimated differential emerges in 2018 and grows over subsequent years, reaching its highest levels toward the end of the observation period. Importantly, the positive dynamics and the absence of systematic differential pre-trends are broadly preserved across the different sample restrictions. Although excluding the top 1% of pre-treatment productivity attenuates the magnitude of the estimated effects, particularly in later years, the overall trajectory remains similar.

Appendix Figure A4 reports the corresponding event-study estimates using the full unbalanced physician panel. The results display a broadly similar dynamic pattern, with no clear systematic divergence before recognition and a progressive increase in publication output thereafter. The persistence of these dynamics across sample restrictions provides further evidence that the main findings are not driven by the balanced-panel requirement.

Appendix Figures A5 and A6 report the corresponding event-study estimates for the probability of publishing at least once in a given year, using the balanced and unbalanced panels, respectively. Unlike the results for publication counts, these estimates reveal no systematic increase in the likelihood of publishing following IRCCS recognition. The coefficients remain relatively small and statistically indistinguishable from zero across most post-treatment periods and sample restrictions. Together with the persistence of positive publication-count effects when restricting the sample to pre-treatment publishers, this pattern suggests that the expansion in scientific output operates primarily through the intensive margin, with already active researchers increasing their publication volume rather than a substantial increase in the proportion of physicians engaged in publishing.

5.3 Effects with Staggered Entry

The Goodman–Bacon decomposition reported in Appendix Figure A7 indicates that approximately 84% of the full-sample TWFE weight comes from comparisons between treated cohorts and physicians who never enter the IRCCS perimeter, while the remaining 16% comes from comparisons among cohorts entering at different times. Both components are positive, although the timing-based comparisons yield a smaller estimated coefficient. To investigate whether the positive response also emerges when identification relies exclusively on variation in the timing of formal entry, we restrict the analysis to physicians who eventually enter the perimeter.

Appendix Figure A8 reports the event-study TWFE estimates from Equation (3), alongside those obtained using the estimators proposed by Callaway–Sant'Anna, Gardner, and Sun–Abraham. Before formal entry, the estimates fluctuate around zero, with some negative coefficients in the years immediately preceding treatment but no pronounced differential trend. Following entry, publication output increases, although the response is initially modest and becomes more substantial over time. By the third year after entry, the estimated increase

ranges from approximately two to three additional publications per physician, depending on the estimator.

The magnitude of the response slightly differs across methods. TWFE and Gardner generally yield larger estimates in the later lost-entry periods, whereas Callaway–Sant'Anna and Sun–Abraham produce more moderate effects. Nevertheless, all four estimators point to a progressive increase in publication output following formal inclusion in the IRCCS perimeter.

6 Robustness

Treatment timing and perimeter composition. We first assess whether the baseline estimates depend on changes in the composition of the IRCCS perimeter. Holding 2018 fixed as the treatment year and retaining never-IRCCS physicians as the comparison group, we separately consider physicians already included at recognition and those entering in later years. Appendix Figure A10 shows a larger post-recognition response among initial entrants, who were also more productive before treatment, but a positive and statistically significant increase among late entrants as well. Appendix Tables A5 and A6 describe these differences and the observable correlates of late entry.

We also address the greater prevalence of academic physicians within the IRCCS perimeter. We first exclude academics from the treated group and then from both the treated and comparison groups. The resulting event studies, reported in Appendix Figure A11, are closely aligned and remain positive after 2018. The baseline effect is therefore not explained by the academic composition of the IRCCS group.[11]

Sensitivity to violations of parallel trends. We apply the procedure developed by Rambachan and Roth (2023), which constructs confidence intervals allowing post-treatment deviations from parallel trends to be related either to the magnitude of pre-treatment deviations or to changes in their slope. Appendix Figure A9 reports the results for the first post-treatment coefficient and for the average effect over 2018–2022.

The first post-treatment effect is more sensitive: under the bounded relative magnitude restriction, its robust confidence interval includes zero at approximately $\overline{M} = 0.9$, while under the smoothness restriction it remains significant over the reported range up to $M = 0.10$. The average post-treatment effect is more robust. Its lower confidence bound remains positive beyond $\overline{M} = 1$ under the relative-magnitude restriction and up to approximately $M = 0.175$ under the smoothness restriction. The average increase in publications therefore survives economically meaningful departures from exact parallel trends.

[11] The comparison restricted to non-academic physicians also makes the groups more similar in their potential exposure to the publication-based Management-by-Objectives scheme introduced in 2017, for which only hospital-employed physicians were eligible.

Influential physicians and cross-group coauthorship. We next progressively exclude physicians in the upper 1%, 5%, 10%, 20%, and 25% of the pooled distribution of average annual publication output during 2012–2017. Because trimming is based exclusively on pre-treatment productivity, post-recognition outcomes do not determine sample inclusion. Appendix Figure A13 shows that the effects decline in absolute terms but remain positive as trimming becomes more restrictive. After excluding the upper 25%, the static estimate is 1.27 additional annual publications, approximately 159% of the corresponding treated-group pre-treatment mean, compared with approximately 70% in the baseline. The result is therefore not confined to the initially most prolific physicians.

We also reconstruct annual output after removing every publication coauthored by IRCCS and non-IRCCS physicians. Appendix Figure A14 shows that the post-recognition coefficients remain positive, although smaller than in the baseline. The 2012 coefficient is positive and statistically different from zero, while the remaining pre-treatment coefficients are smaller and do not reveal a persistent monotonic trend. Cross-group collaboration thus contributes to the measured expansion but does not fully account for it.

Outcome definitions and additional checks. The post-recognition increase is also preserved when annual publications are transformed using $\log(1 + Y_{it})$ or the inverse hyperbolic sine (Appendix Figure A15). Finally, we reconstruct publication counts after excluding COVID-19-related articles and separately exclude physicians observed leaving the IRCCS perimeter after entry. Appendix Figure A12 shows that removing pandemic-related publications leaves the trajectory largely unchanged, while excluding physicians who opt out produces somewhat larger post-recognition coefficients. Overall, the main dynamic pattern is robust across alternative treatment-group definitions, sample restrictions, outcome transformations, and publication exclusions.

7 Heterogeneity

The previous section established that IRCCS recognition increased scientific output on average. We now examine whether physicians' responses depend on their pre-existing research capacity. Physicians are classified into terciles according to their average annual publication output over 2012–2017, and the baseline event-study specification is estimated separately within each group. Figure 3 reports the resulting estimates.

This heterogeneity is informative about the organizational channels through which additional capacity translates into scientific production. In their analysis of research laboratories, Carayol and Matt (2004) show that publication performance depends on the combination of researchers and supporting inputs, rather than on individual characteristics alone. Relatedly, Heinze et al. (2009) document how access to complementary skills, research support,

and external resources contributes to environments conducive to scientific creativity. These contributions motivate examining whether the new IRCCS environment disproportionately benefits physicians with the experience and established relationships needed to mobilize such inputs. Pre-treatment publications provide an observable proxy for existing research activity, not a direct measure of individual ability or absorptive capacity.

The post-recognition response displays a clear gradient with respect to prior scientific activity. Publication output increases most strongly among physicians in the upper tercile, with the difference emerging shortly after recognition and widening over the subsequent years. Physicians in the middle tercile also experience a sustained increase, although smaller in magnitude. The response in the bottom tercile is more limited and develops more gradually.

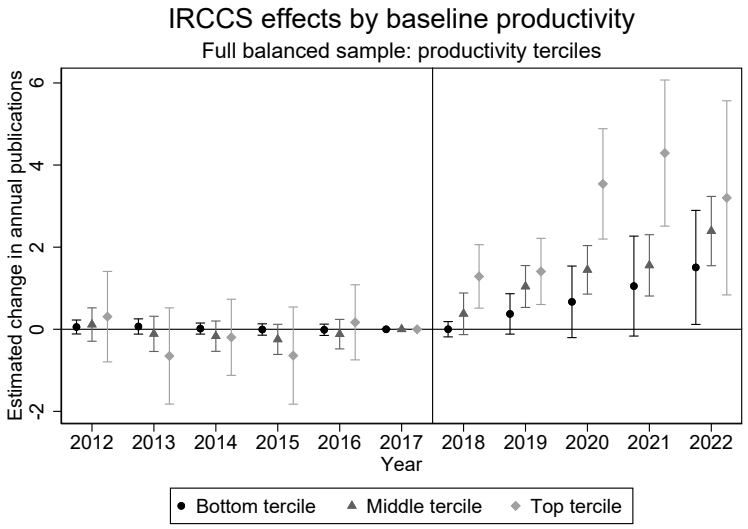


Figure 3: Effect of IRCCS recognition by baseline productivity. Physicians are classified into terciles according to their average annual publication output over 2012–2017. The figure reports separate event-study estimates for each tercile using the balanced physician panel. All specifications include physician and department-by-year fixed effects. The omitted reference year is 2017. Vertical bars report 95% confidence intervals based on standard errors clustered at the physician level.

Viewed against this literature, the productivity gradient is consistent with organizational complementarities accruing disproportionately to physicians already engaged in research. It does not establish which specific input or mechanism accounts for these heterogeneous treatment effects, nor does it imply that the institution selected the most intrinsically able researchers.

The expansion in scientific production was therefore not confined exclusively to physicians who were already the most productive before recognition. Nevertheless, the substantially larger response in the upper part of the distribution indicates that pre-existing research capacity shaped physicians' ability to benefit from the new institutional environment. Existing scientific experience, research networks, and organizational familiarity may have allowed

more research-active physicians to translate additional resources and support into publications more rapidly and on a larger scale. In this sense, the response to IRCCS recognition was broad but capability-biased in magnitude.

Because the bottom tercile includes physicians with no publications during the pre-treatment period, Appendix Figure A16 repeats the analysis after restricting the sample to physicians with at least one publication over 2012–2017 and recalculating the productivity terciles within that sample. The post-recognition gradient remains visible, although the estimates—particularly those for the most productive publishers—are less precise and display greater pre-treatment variability.

8 Research Scaling or Transformation?

The preceding analysis shows that IRCCS recognition substantially increased scientific production. We now examine whether this increase primarily reflects an expansion of existing research activity or a broader transformation of the institution's scientific portfolio, collaborative organization, and research impact. We first study the distribution of publications across scientific fields, then examine changes in collaboration and authorship patterns, and finally assess whether greater publication output translated into greater scientific impact.

8.1 Research Portfolio

We begin by examining whether IRCCS recognition altered the scientific orientation of research activity or primarily expanded production within established fields. Using Web of Science subject classifications, we construct physician-year measures of publication output across broad clinical and scientific areas. The original subject categories are aggregated into macro-specializations according to the correspondence reported in Appendix Table A7.

Research specialization is an organizational margin in its own right. Heinze et al. (2009) emphasize the importance of access to heterogeneous expertise and opportunities for scientific exchange across cognitive boundaries, including instances in which highly creative scientists move between research fields. Carayol and Matt (2004) similarly show that laboratories differ in their disciplinary composition and in how personnel inputs are combined. These findings motivate looking beyond the number of publications in each field: a change in field shares would be consistent with a reorientation of research activity, whereas growth within largely stable shares would indicate the expansion of established specializations. Neither measure, however, directly identifies interdisciplinary exchange or scientific novelty.

Table 3 reports separate difference-in-differences estimates for field-specific publication output. Given the relatively large number of outcomes, we report both conventional and

Romano–Wolf stepdown adjusted p-values (Romano & Wolf, 2005, 2016), alongside joint tests of the pre-treatment event-study coefficients.

The estimates indicate that the increase in scientific production extends across several research areas rather than being concentrated in a single specialization. Particularly pronounced increases emerge in Internal Medicine, Biology, Infectious Diseases and Immunology, Cardiology, Public Health, and Pharmacology. Many of these effects remain statistically significant after adjusting for multiple hypothesis testing. Positive effects are also observed in Oncology and Hematology, Diagnostics and Imaging, and Maternal and Child Health, although their pre-treatment dynamics warrant greater caution. The corresponding field-specific event studies are reported in Appendix Figures A17–A19.

However, greater publication output across multiple fields does not necessarily imply a reorientation of the hospital's scientific portfolio. To investigate this distinction, Figure 4 presents the evolution of the hospital-wide distribution of scientific activity across macro-specializations over 2012–2022. The figure counts each Web of Science publication only once and expresses field-specific activity as a share of total macro-field mentions, allowing publications to contribute to multiple scientific areas.

Despite the substantial expansion in scientific output, Figure 4 shows that the relative distribution of research activity across fields remained broadly stable around recognition. Some field shares fluctuate over time, but there is no apparent systematic reallocation of the hospital's scientific portfolio toward new areas. The increase in publication volume therefore appears to have occurred largely within the institution's existing areas of specialization.

We complement this hospital-level descriptive evidence with physician-level difference-in-differences estimates of publication shares across macro-specializations. These results, reported in Appendix Table A8, provide little evidence of systematic reallocation. Estimated changes in field shares are generally small, and none remains statistically significant after Romano–Wolf adjustment. The corresponding event studies in Appendix Figures A20–A22 similarly reveal no clear structural break in portfolio composition.

In relation to Heinze et al. (2009), the absence of a measurable portfolio shift is particularly informative because the organizational availability of diverse expertise need not result in a reallocation of publication output across scientific fields. Our results capture the composition of realized scientific production, not the diversity of ideas or methods exchanged within projects. They therefore indicate continuity in specialization without ruling out intellectual exchange within those established fields.

Taken together, the evidence suggests that IRCCS recognition expanded scientific production across several established areas of expertise without substantially redirecting the institution's research portfolio. The primary response was therefore a scaling-up of existing scientific activity rather than a pronounced transformation of its disciplinary composition.

Table 3: Effect of IRCCS recognition on scientific output by field

Field	Coef.	SE	p-value	RW p-value	Pre-test	Treated pre-mean	Obs.
Biology	0.233***	(0.056)	0.000	0.002***	0.575	0.337	6,421
Cardiology/Cardiovasc. Diseases	0.159***	(0.058)	0.006	0.015**	0.223	0.365	6,421
Chemistry and Physics	0.044**	(0.020)	0.031	0.033**	0.541	0.049	6,421
Diagnostics and Imaging	0.140***	(0.045)	0.002	0.007***	0.114	0.280	6,421
Infectious Dis./Immunology	0.263***	(0.050)	0.000	0.001***	0.070	0.163	6,421
Internal Medicine (Generic)	0.482***	(0.069)	0.000	0.001***	0.532	0.236	6,421
Internal Medicine (Specialties)	0.224***	(0.076)	0.003	0.011**	0.015	0.528	6,421
Maternal and Child Health	0.229***	(0.064)	0.000	0.007***	0.319	0.244	6,421
Neurosciences and Psychology	0.078	(0.057)	0.173	0.253	0.818	0.477	6,421
Oncology and Hematology	0.391***	(0.102)	0.000	0.005***	0.035	0.501	6,421
Pharmacology	0.159***	(0.030)	0.000	0.001***	0.184	0.126	6,421
Public Health and Health Systems	0.177***	(0.036)	0.000	0.001***	0.380	0.064	6,421
Rehabilitation and Nursing	-0.010	(0.020)	0.612	0.548	0.161	0.035	6,421
Social Sciences and Humanities	0.008	(0.011)	0.470	0.548	0.443	0.021	6,421
Surgery and Procedures	0.181***	(0.054)	0.001	0.007***	0.370	0.343	6,421
Technology and Engineering	-0.042	(0.043)	0.331	0.490	0.017	0.161	6,421

8.2 Research Organization

We next examine whether the increase in publication output was accompanied by changes in the organization of scientific production. We consider two dimensions: the development of physicians' collaboration networks and their positions within research teams.

These margins follow directly from the organizational literature. Carayol and Matt (2004) conceptualize scientific output as a collective production process in which the mix of personnel and complementarities within research units matter for performance. Heinze et al. (2009) identify communication across scientific and organizational boundaries, access to external expertise, and facilitating leadership as features of creative research environments. We accordingly distinguish growth in collaboration intensity from the formation of new ties and from collaboration across institutional or professional categories; we separately use authorship positions to describe changes in researchers' roles within teams.

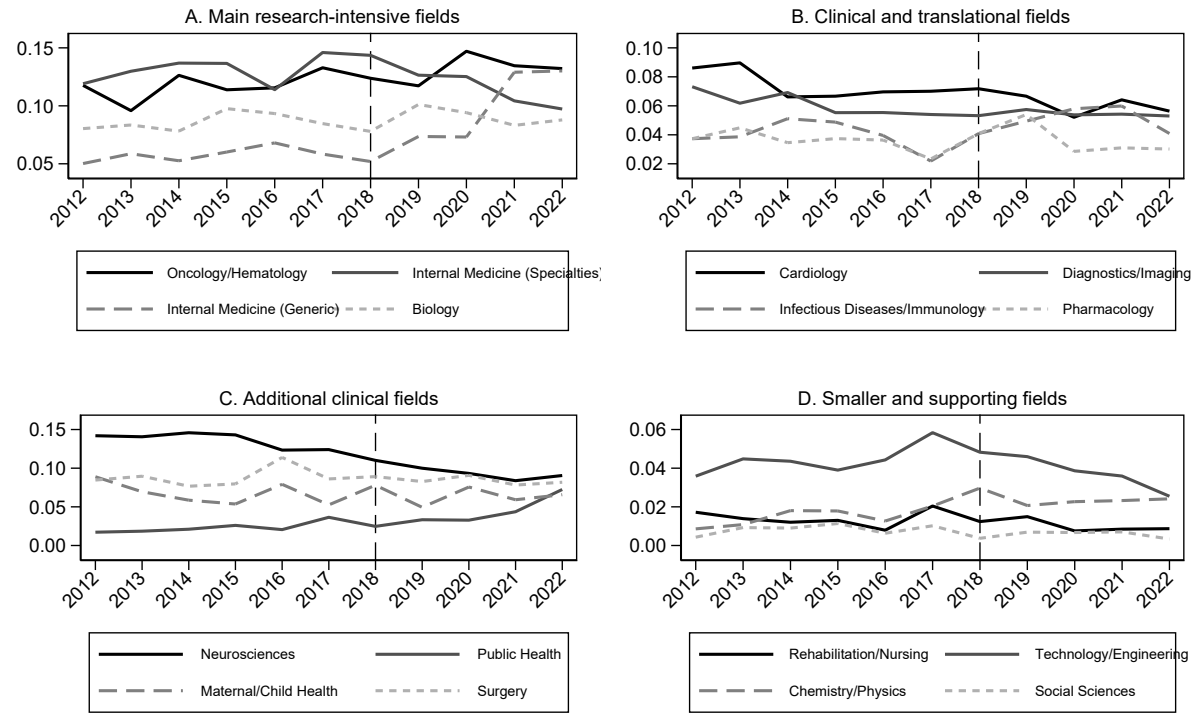


Figure 4: Hospital research portfolio by scientific field. The figure reports annual shares of Web of Science macro-field mentions among unique publications associated with physicians in the balanced panel. Publications may be assigned to multiple macro-fields. The vertical dashed line indicates IRCCS recognition in 2018.

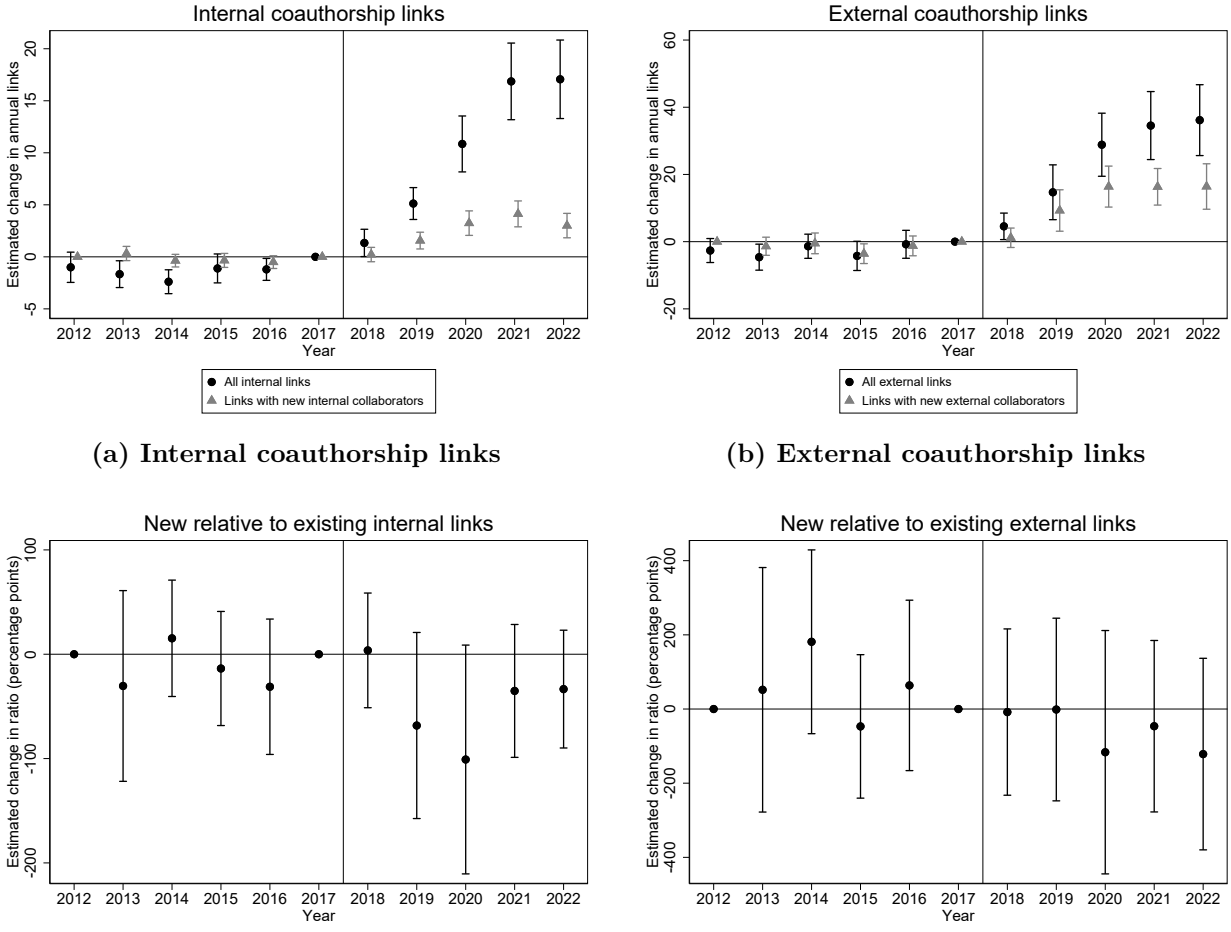
Collaboration networks

For each physician-year, we count coauthorship links with researchers inside and outside the hospital. A link is classified as internal when the coauthor is observed among the hospital's physicians in the same year and as external otherwise. We distinguish total coauthorship links from links involving a collaborator with whom the physician had not previously published during the observation period. These outcomes capture collaboration intensity: repeated joint publications with the same coauthor contribute multiple links.

Figure 5 reports the event-study estimates for the balanced physician panel. Both internal and external collaboration increase markedly after IRCCS recognition. The response is initially limited but becomes substantial from 2019 onward. By 2021–2022, treated physicians record roughly 15–17 additional annual internal links and more than 30 additional external links relative to non-IRCCS physicians.

Links involving new collaborators also increase after recognition. The increase is visible

for both internal and external relationships, although it is considerably smaller than the increase in total links. The expansion in scientific production therefore involved both the formation of new collaborations and a stronger intensification of relationships with existing coauthors. The larger response in external links further indicates that the increase in research activity was not confined to collaborations within the hospital.



(a) Internal coauthorship links

(b) External coauthorship links

(c) New-to-existing internal links

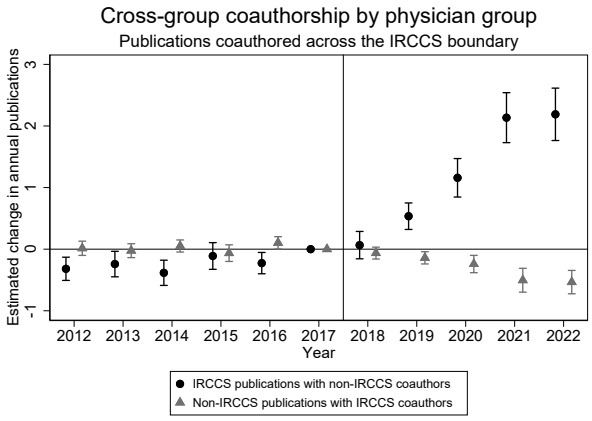
(d) New-to-existing external links

Figure 5: IRCCS recognition and collaboration intensity. Panels (a)–(b) report event-study estimates for annual internal and external coauthorship links; Panels (c)–(d) report ratios of new to previously observed links. Internal coauthors are physicians observed in the hospital in the same year. Ratio outcomes exclude 2012. All specifications use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

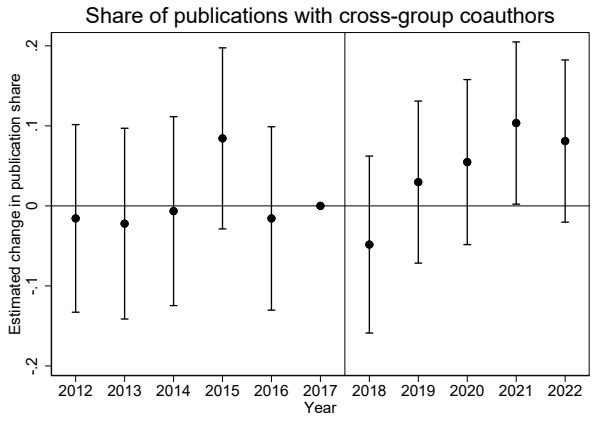
The distinction between new and established ties is related to the contrasting network mechanisms discussed by Heinze et al. (2009): cohesive relationships may ease knowledge exchange and coordination, while connections across otherwise separate groups may provide access to different knowledge sources. Total coauthorship links alone cannot distinguish these organizational responses, since a rise in publication volume mechanically creates more observed coauthorship links. The relative importance of new ties provides an additional,

albeit imperfect, indication of whether collaborative production is being reorganized.

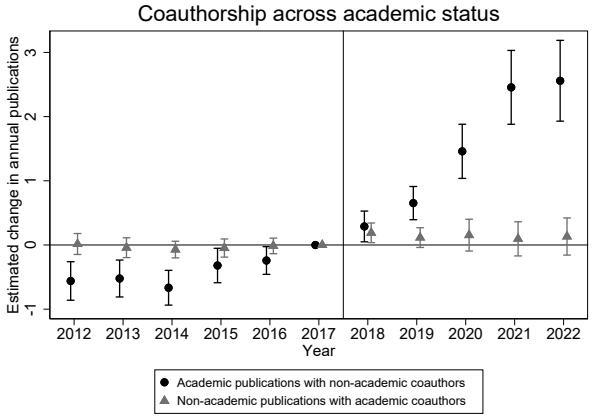
Panels (c) and (d) of Figure 5 report the corresponding ratios of new to existing links. These estimates are imprecise and do not show a systematic post-recognition increase. Thus, although treated physicians formed additional relationships, the evidence does not indicate that new collaborators displaced established ones within their collaboration portfolios. Estimates based on distinct collaborators are reported in Appendix Figure A23. These estimates are noisier and display some pre-treatment differences, so we treat them as supporting evidence rather than as a separate identification result.



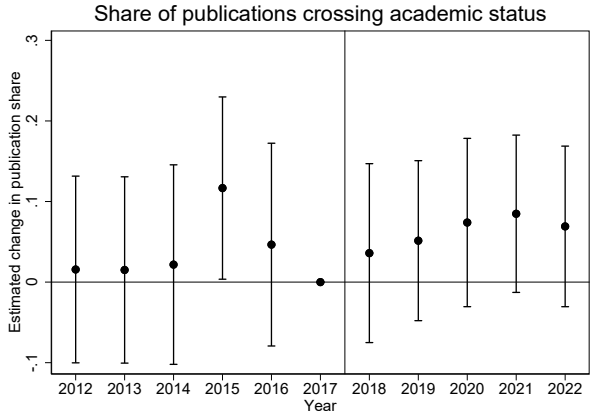
(a) Coauthorship across the IRCCS perimeter



(b) Share across the IRCCS perimeter



(c) Coauthorship across academic status



(d) Share across academic status

Figure 6: Collaboration across organizational boundaries. Panels (a)–(b) report counts and shares of publications crossing the IRCCS perimeter; Panels (c)–(d) report the corresponding outcomes across academic status. Shares are defined for physician-years with positive output. Each paper is counted once per focal physician. All specifications use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

Cross-boundary collaborations are another potential source of complementary expertise. Carayol and Matt (2004) find associations between laboratories' publication performance and their mix of researchers holding different professional roles, while Heinze et al. (2009)

underline the value of timely access to skills and resources beyond the immediate research group. This motivates studying links not only within and outside the hospital, but also across the IRCCS research perimeter and between academic and non-academic physicians. Such boundaries are administrative proxies for potentially different research roles and resources, rather than measures of cognitive distance.

We next examine collaboration across the formal IRCCS boundary. Figure 6 distinguishes publications of IRCCS physicians coauthored with internal non-IRCCS physicians from publications of non-IRCCS physicians coauthored with internal IRCCS physicians. Panel (a) reveals a marked asymmetry between the two groups. Among IRCCS physicians, cross-group publications increase substantially after recognition, reaching approximately two additional publications per year by 2021–2022. In contrast, non-IRCCS physicians exhibit no corresponding increase in publications involving IRCCS coauthors, with estimates becoming slightly negative in the later years.

However, Panel (b) provides little evidence of a systematic change in the share of publications involving cross-group coauthorship. Although the estimated coefficients become positive toward the end of the observation period, they remain imprecisely estimated and statistically indistinguishable from zero. Taken together, these findings suggest that IRCCS physicians expanded their scientific production partly through collaborations with colleagues outside the formal research perimeter, without a corresponding increase in cross-group collaboration among non-IRCCS physicians or a clear reallocation of publication activity toward such relationships.

Panels (c)–(d) of Figure 6 also examine publications connecting academic and non-academic physicians. This comparison is motivated by the complementarity between different researcher roles highlighted by Carayol and Matt (2004). Academic affiliation and clinical employment in our hospital, however, do not correspond exactly to the full-time-researcher and professor categories in their university laboratory setting. Accordingly, these outcomes test for observable changes in cross-status collaboration, not for the productivity effect of a particular optimal staffing mix. In interpreting the figure, changes in the number of cross-status publications should be distinguished from changes in their share of total publication output.

Positions within research teams

We next examine whether expanded collaborative production changed physicians' positions within the authorship hierarchy. We separately count first-, middle-, and last-authored publications. In biomedical research, first authorship generally indicates primary responsibility for a study, while last authorship often reflects senior scientific supervision. Middle authorship captures participation in broader collaborative teams.

The distinction also connects to the role of leadership in scientific organizations. Heinze et al. (2009) discuss how research leaders facilitate access to resources, help coordinate scientific work, and sustain productive research environments. Their concept of leadership is organizational and cannot be directly recovered from bibliometric author order. Nevertheless, first-, middle-, and last-authorship positions offer imperfect indicators of whether added research capacity expands individual study leadership, senior supervision, or participation in collective projects. In particular, growth in last-authored work may be consistent with more supervisory activity but should not be equated with a change in the quality or distribution of scientific leadership.

Figure 7 shows that the increase in scientific output is concentrated primarily in middle-authored publications. The effect appears soon after recognition and grows sharply over time, reaching approximately two additional middle-authored publications per year in 2021 and 2022. This pattern is consistent with treated physicians participating in a growing number of collaborative research projects.

Last-authored publications also increase progressively after recognition, although by a smaller amount. This result suggests some expansion in senior or supervisory research activity alongside the broader increase in team participation. By contrast, first-authored publications do not exhibit a clear post-recognition break: their coefficients are modest and similar in magnitude before and after 2018.

Taken together, these estimates indicate that the increase in publication output was associated primarily with greater participation in collaborative teams and, to a lesser extent, with additional senior authorship. The results do not support a generalized shift toward first-authored production.

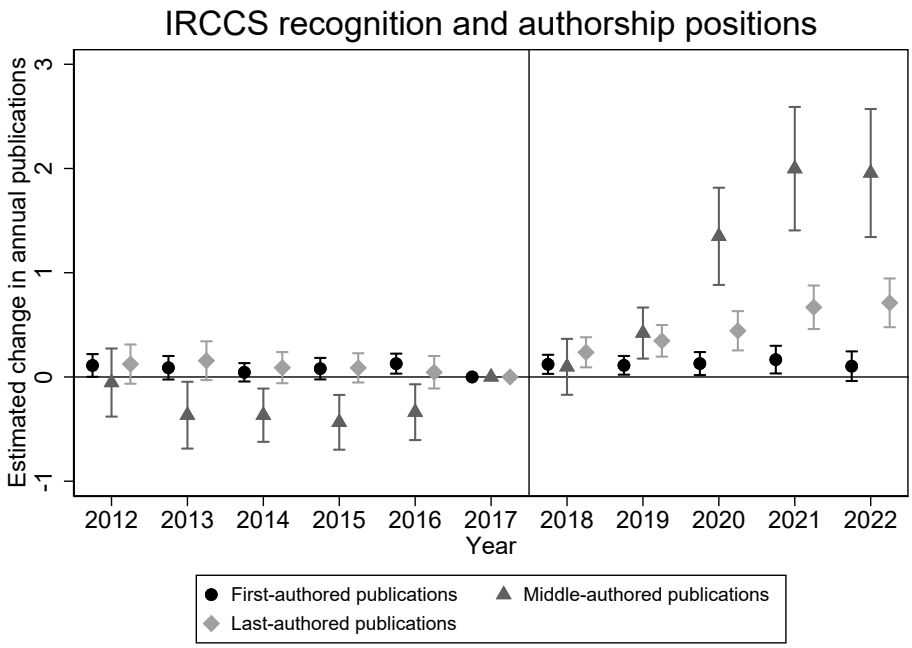


Figure 7: IRCCS recognition and authorship positions. Event-study estimates for annual first-, middle-, and last-authored publications using the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

Thus, the prominence of middle authorship is consistent with the collective-production perspective of Carayol and Matt (2004), while the more modest increase in last authorship is compatible with some expansion of supervisory research roles. Unlike the creative research cases studied by Heinze et al. (2009), these estimates do not identify whether scientific leadership became more facilitating, or whether research became more exploratory. They indicate how the volume of published work is allocated across observable team positions.

Finally, Appendix Figure A24 examines the average number of coauthors per publication. The estimates are highly imprecise and do not show a persistent post-recognition increase. The organizational response therefore appears to operate through participation in a larger number of collaborative projects, rather than through a clearly detectable increase in the average size of each individual publication team.

8.3 Scientific Impact

Finally, we examine whether the increase in publication output translated into greater citation impact. Citations are observed in October 2024, so publications from different years have had different amounts of time in which to accumulate citations. We therefore annualize each publication's citation count by dividing it by the number of years elapsed between its publication year and 2024.

We consider two outcomes. The first is the total number of annualized citations generated by each physician's publications in a given year. This measure captures aggregate citation impact and assigns zero to physician-years without publications. The second is annualized citations per publication, calculated only for physician-years with positive publication output. It captures average citation impact among observed publications but conditions on publishing, which may itself respond to IRCCS recognition.

The event studies exhibit statistically significant differences during the pre-treatment period. We therefore report the estimates together with the pre-trend power diagnostic proposed by Roth (2022). For each outcome, the procedure calculates the magnitude of a linear differential trend that the conventional pre-trend test would detect with 50% probability and projects that trend into the post-treatment period.

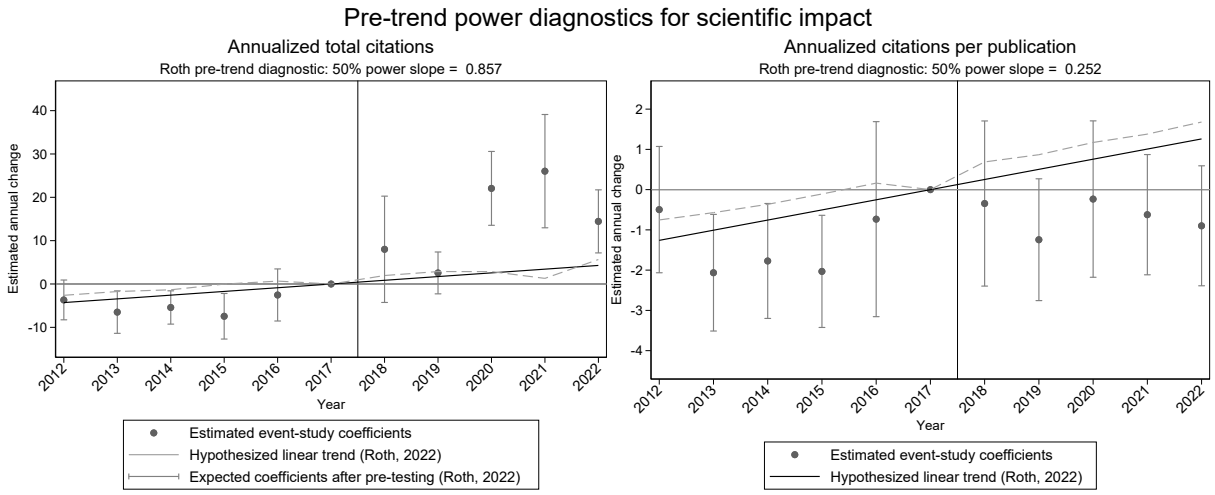


Figure 8: IRCCS recognition and scientific impact: pre-trend diagnostics. Markers show event-study estimates with 95% confidence intervals; solid and dashed lines report the detectable linear trend and conditional expected coefficients from Roth (2022). Outcomes are annualized total citations and citations per publication, measured in October 2024. Specifications use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is omitted.

Aggregate citation impact increases substantially after IRCCS recognition. The effect is initially modest but becomes large in 2020 and 2021, when the estimated increase exceeds 20 annualized citations per physician. These post-treatment estimates are substantially larger than the continuation of the linear differential trend represented in the Roth diagnostic. The growth in aggregate citation impact is therefore not mechanically accounted for by that particular extrapolation of the pre-treatment differences.

Annualized citations per publication, by contrast, do not exhibit a comparable post-recognition increase. The coefficients remain generally negative relative to 2017, are imprecisely estimated, and lie below the projected linear trend. The corresponding static difference-in-differences coefficient is also not statistically distinguishable from zero. The

evidence therefore does not indicate that the expansion in publication output was accompanied by an increase in average citation impact per article.

These findings require a cautious interpretation. The pre-treatment coefficients are jointly significant for both outcomes, so the citation estimates do not provide the same clean difference-in-differences evidence as the baseline publication results. Moreover, citations per publication are observed only among publishing physician-years, and publication itself responds to treatment. We therefore interpret this analysis as showing that the expansion in scientific production was accompanied by greater aggregate citation reach, while the data do not establish either an improvement or a causal deterioration in average impact per publication.

8.4 Clinical Activity and Outcomes

We complement the physician-level analysis with hospital-level indicators from the National Outcomes Programme (PNE) to examine whether the expansion in scientific production coincided with changes in clinical activity and patient outcomes. We compare the treated hospital with non-IRCCS hospitals in Lazio over 2015–2022, focusing on six high-complexity or technology-intensive procedure volumes and six 30-day mortality indicators. The hospital fixed-effects estimates, event studies, and pre-trend diagnostics are reported in Appendix Figures A25 and A26 and Appendix Tables A9 and A10.

The evidence points to substantial increases in selected complex clinical activities after recognition. Estimated changes in procedure volumes are positive for all six outcomes, including minimally invasive lung and colon cancer surgery, pancreatic cancer resections, and carotid endarterectomies. In contrast, 30-day mortality indicators display mixed movements, with no consistent improvement in measured clinical outcomes. These patterns suggest that the expansion in scientific production coincided with greater activity in some technologically demanding clinical areas, but not with a systematic improvement in patient outcomes. This evidence is descriptive rather than causal: the analysis concerns a single treated hospital, the pre-treatment period is short, and several outcomes exhibit differential pre-trends. Changes in patient composition, referrals, or procedure selection may also contribute to the observed patterns.

9 Conclusions

This paper asks whether an expansion in organizational research capacity produces more science along established trajectories or transforms the way science is organized and directed. We study the recognition of a large Italian university hospital as an IRCCS institute, a change that provided recurring public funding, infrastructure, formal research lines, and

organizational support. The intervention is therefore best understood as a change in the institution's research-production capacity rather than as an individual grant program.

Three results characterize the response. First, exposure to the IRCCS research environment produces a large and persistent increase in publications. Second, the response is capability-biased: physicians who were already research-active and more productive before recognition expand the most. Third, the additional output is produced mainly within the hospital's established scientific architecture. Existing collaboration networks become more active, output grows across familiar research fields without a comparable reallocation of portfolio shares, and authorship patterns provide little evidence of a broad shift toward new scientific leadership. Total citations increase with publication output, while we find no robust evidence of an improvement in citations per publication.

The pattern also helps position our findings relative to the organizational literature. Whereas Carayol and Matt (2004) establish that team composition and complementary labor inputs are associated with differences in laboratory productivity, we observe how scientific output and collaboration respond to a change in institutional research capacity within a hospital. Heinze et al. (2009) identify organizational conditions associated with creative breakthroughs, including the availability of diverse expertise, external resources, and facilitating leadership. Our evidence speaks to several observable organizational margins emphasized by that study, but it does not test whether IRCCS recognition caused scientific creativity or breakthrough research.

The evidence therefore points to scaling rather than transformation. This conclusion does not rest on any single insignificant coefficient. It follows from the coherence of the margins: initial capability predicts the size of the response; existing teams absorb much of the expansion; new ties do not become relatively more important; and the scientific portfolio remains stable. Additional capacity is productive, but it is deployed through the routines, relationships, and areas in which the organization already possesses absorptive capacity.

These findings have two implications for research policy. First, publication counts alone provide an incomplete assessment of capacity-building policies. Evaluations should distinguish additional output from changes in the direction and organization of knowledge production. Second, funding, infrastructure, and organizational support need not be sufficient to diversify science. If the policy objective includes new research domains, cross-field integration, or new scientific leadership, capacity expansion may need to be paired with instruments that directly support entry, exploration, and collaboration beyond established networks.

Our setting is one healthcare organization, and IRCCS recognition bundles several inputs that cannot be separately randomized. The estimates should accordingly be interpreted as the intention-to-treat effect of exposure to a new institutional research environment, not as the marginal return to a specific euro or funding component. The single-organization setting is

also an advantage for observing the internal process of scientific expansion in unusual detail. Future work should examine whether capability-biased scaling generalizes to universities, public laboratories, and other mission-oriented research organizations, and whether targeted instruments can turn capacity growth into durable scientific diversification.

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Appendix

Requirement	**Statutory criterion**	**Evidence required**
a. Legal status	*Public or private legal personality and valid authorization and accreditation to provide healthcare* (Art. 13(3)(a)–(b))	• Acts proving legal personality • Healthcare authorization/accreditation • Number of accredited beds • Relevant healthcare specialties
b. Capacity	*Efficient organization, adequate facilities and technological equipment, and financial sustainability* (Art. 13(3)(c))	• Size of healthcare/research facilities • Organizational structure • Personnel by contract type/profile • Personnel dedicated to research • Agreements with unis/research institutes • Financial statements (3y) • Separate research accounts (3y) • Economic and efficiency indicators
c. Clinical excellence	*Excellence in highly specialized healthcare activity in the area for which recognition is requested* (Art. 13(3)(d))	• Number of beds • Annual admissions • Admissions for relevant pathologies • Average length of stay • Bed occupancy rate • Comparative performance index • Case-mix index • Interregional patient attraction • Significant/innovative equipment (care) • Clinical pathways • Innovative procedures • Clinical trials • Disease registries • Biological material banks
d. Scientific excellence	*Excellence in research activity during the previous three years in the field for which recognition is requested* (Art. 13(3)(e))	• Normalized IF; Norm. IF per person • Sign./innov. equipment (research) • GMP/other laboratories • Funded and active research projects • Clinical trials; enrolled patients • Laboratories/services for research • Research lines • Patents • Translational research products • Scientific articles • Biological material banks • Internal research training projects
e. Research networks	*Demonstrated ability to participate in research networks and collaborate with other public and private institutions* (Art. 13(3)(f))	• Digitalization of healthcare activities • Broadband availability • Videoconferencing equipment • Network applications • Services provided through networks • Regional /national reference centers • Foreign collborations
f. Ext. research funding	*Demonstrated ability to attract independent public and private research funding* (Art. 13(3)(g))	• Public contributions • Private contributions
g. Quality certification	*Certification of service quality according to internationally recognized procedures* (Art. 13(3)(h))	• ISO certification • Joint Commission certification • OECI certification

Table A1: Main requirements for IRCCS recognition and supporting evidence required from applicant institutions. The statutory criteria reported in the second column summarize the requirements set out in Art. 13(3) of Legislative Decree 288/2003, as applicable at the time of the hospital's recognition. The documentary evidence reported in the third column is drawn from the documentation requirements established by the Ministerial Decree of 5 February 2015. These were the legal and documentary requirements referenced in the hospital's 2018 recognition decree.

Step	Description
1. Regional submission	On 10 October 2017, the Lazio Region transmitted the hospital's application to the Ministry of Health and certified its consistency with regional health planning.
2. Evaluation commission	On 14 November 2017, the Ministry of Health appointed an expert commission to assess the application and verify the requirements for recognition.
3. First site visit	On 21 December 2017, the ministerial evaluation commission conducted the first on-site assessment of the hospital.
4. Second site visit	On 8 January 2018, the commission conducted a second on-site assessment as part of the recognition procedure.
5. Institutional approval	In February 2018, the application received the required favorable opinions from the competent national and regional authorities.
6. Formal recognition	On 28 February 2018, the Ministry of Health formally recognized the hospital as an IRCCS in the two complementary disciplines of Personalized Medicine and Innovative Biotechnologies.
7. Periodic reassessment	Following recognition, IRCCS status remained subject to subsequent verification of the scientific, clinical, and organizational requirements underlying the designation.

Table A2: Main steps in the hospital's IRCCS recognition process, as documented in the ministerial recognition decree and the applicable recognition procedure.

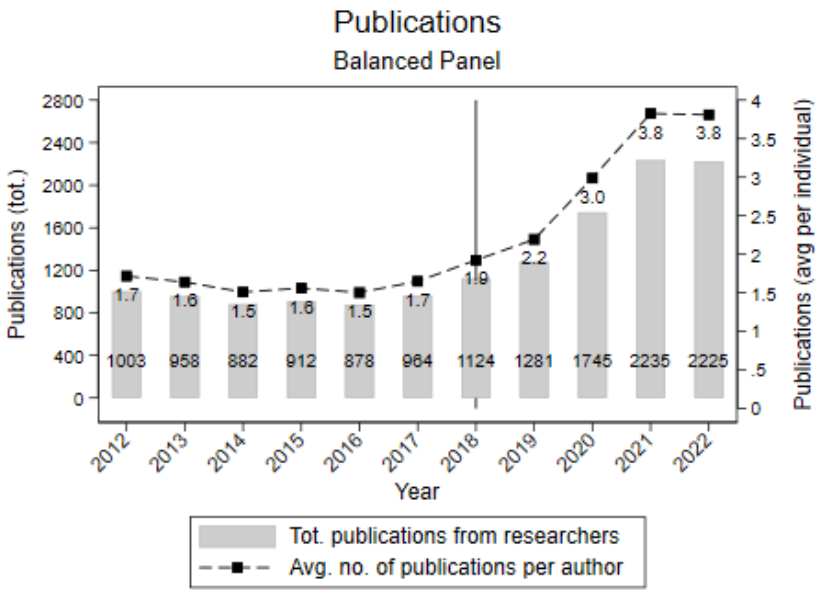


Figure A1: Total number of unique publications attributed to employees of the institution in the balanced sample (2012–2022). Publications co-authored by multiple sampled physicians are counted only once in the bars. The line reports the corresponding average number of publications per researcher, for which a co-authored paper contributes to each author's individual publication count.

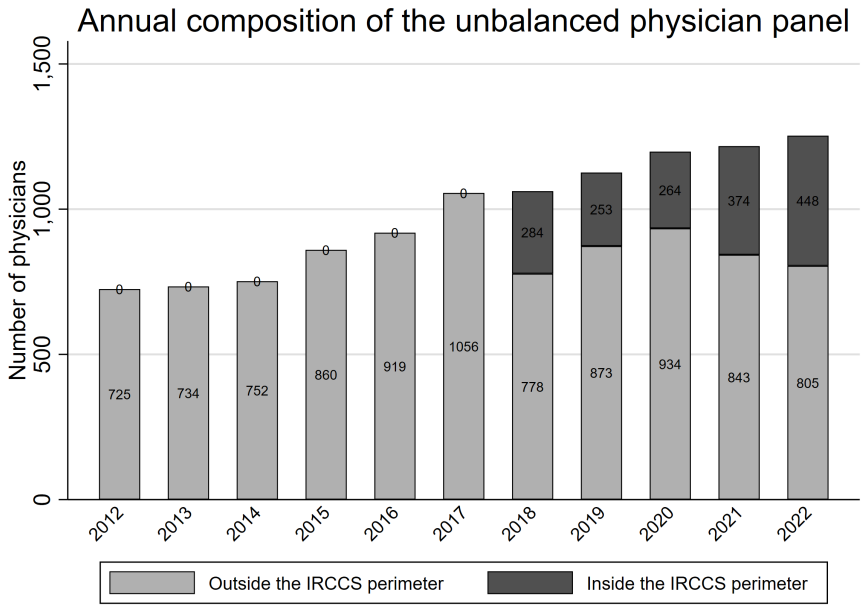


Figure A2: Annual composition of the unbalanced physician panel. The figure reports the number of physicians observed in each year of the full unbalanced panel, distinguishing physicians formally included in the IRCCS research perimeter in that year from physicians observed outside the perimeter. Formal membership is measured using the time-varying IRCCS-perimeter indicator.

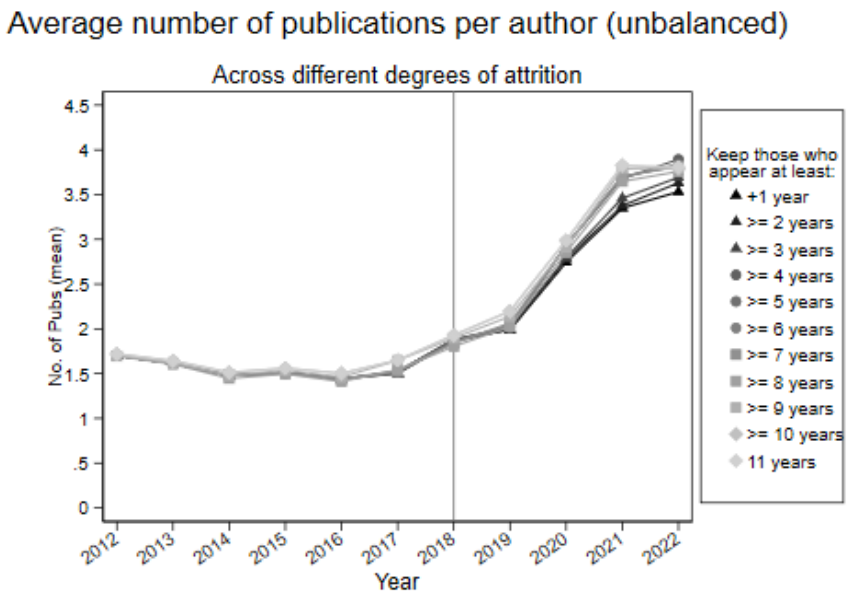


Figure A3: Average publications per researcher per year (2012–2022) across alternative degrees of panel attrition. Each series restricts the sample to physicians observed for at least a given number of years.

	IRCCS			Non-IRCCS			Total		
	Mean	SD	N°	Mean	SD	N°	Mean	SD	N°
Annual publications	3.49	(3.38)	2,288	0.98	(1.49)	2,123	2.28	(2.92)	4,411
Citations (Web of Science)	95.36	(205.40)	2,288	17.78	(41.63)	2,123	58.02	(155.62)	4,411
Citations (all outlets)	101.18	(215.97)	2,288	18.95	(44.16)	2,123	61.61	(163.76)	4,411
Age	51.79	(7.96)	2,288	50.90	(8.05)	2,123	51.36	(8.01)	4,411
Female	0.39	(0.49)	2,288	0.42	(0.49)	2,123	0.41	(0.49)	4,411
Tenure	16.94	(10.21)	2,288	16.17	(10.12)	2,123	16.57	(10.17)	4,411
Co-authors per paper	8.77	(7.18)	2,288	4.43	(6.72)	2,123	6.68	(7.29)	4,411

Table A3: Descriptive statistics for the restricted sample excluding physicians with no pre-treatment publications and physicians in the upper tail of the publication distribution.

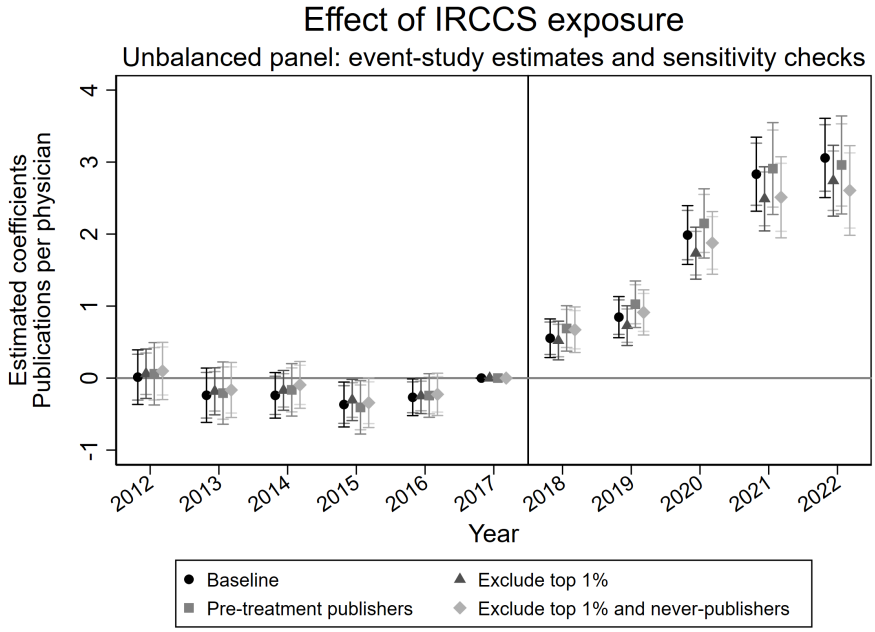


Figure A4: IRCCS recognition and scientific production: dynamic effects. The figure reports event-study estimates of the effect of IRCCS recognition on annual publication counts, comparing physicians ever included in the IRCCS research perimeter with those never included. Estimates are based on the unbalanced physician panel (2012–2022). The baseline specification is compared with three sample restrictions: excluding the top 1% of physicians by average annual pre-treatment publication output (2012–2017), retaining only physicians with at least one pre-treatment publication, and combining both restrictions. All specifications include physician and department-by-year fixed effects. Standard errors are clustered at the physician level. The reference year is 2017, with its coefficient normalized to zero. Vertical bars report 90% and 95% confidence intervals. The vertical line marks IRCCS recognition in 2018.

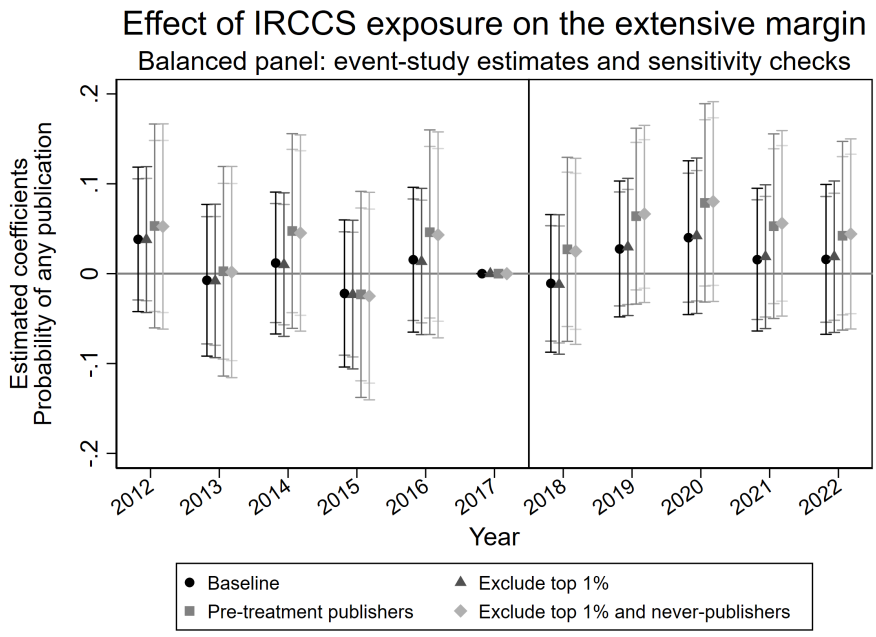


Figure A5: Event-study estimates for the probability of producing at least one publication in a given year using the balanced physician panel. The figure reports the same four sample specifications as the publication-count event study. All specifications include physician and department-by-year fixed effects. Standard errors are clustered at the physician level. The omitted year is 2016.

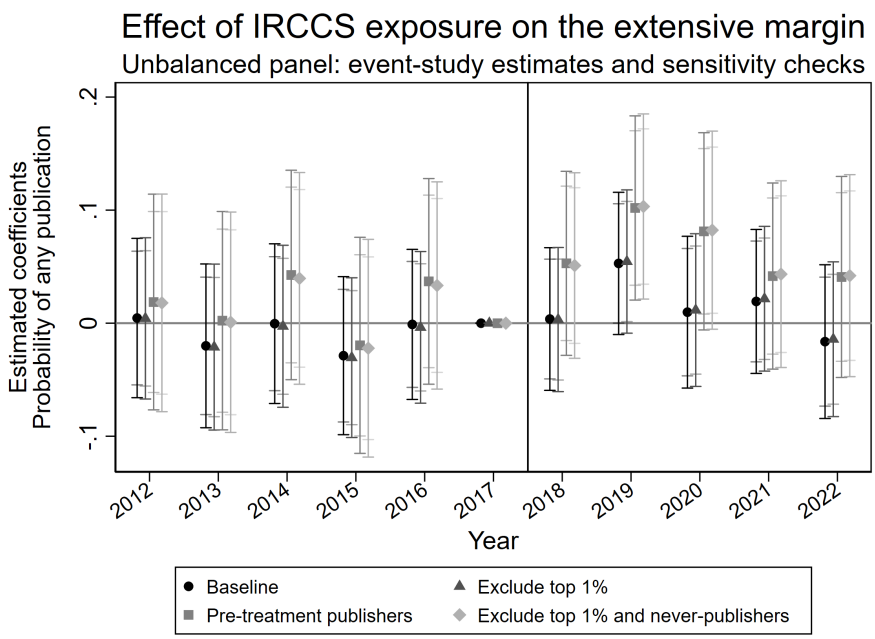


Figure A6: Event-study estimates for the probability of producing at least one publication in a given year using the full unbalanced physician panel. The figure reports the baseline specification and the same three sample restrictions used in the main event-study analysis. All specifications include physician and department-by-year fixed effects. Standard errors are clustered at the physician level. The omitted year is 2016.

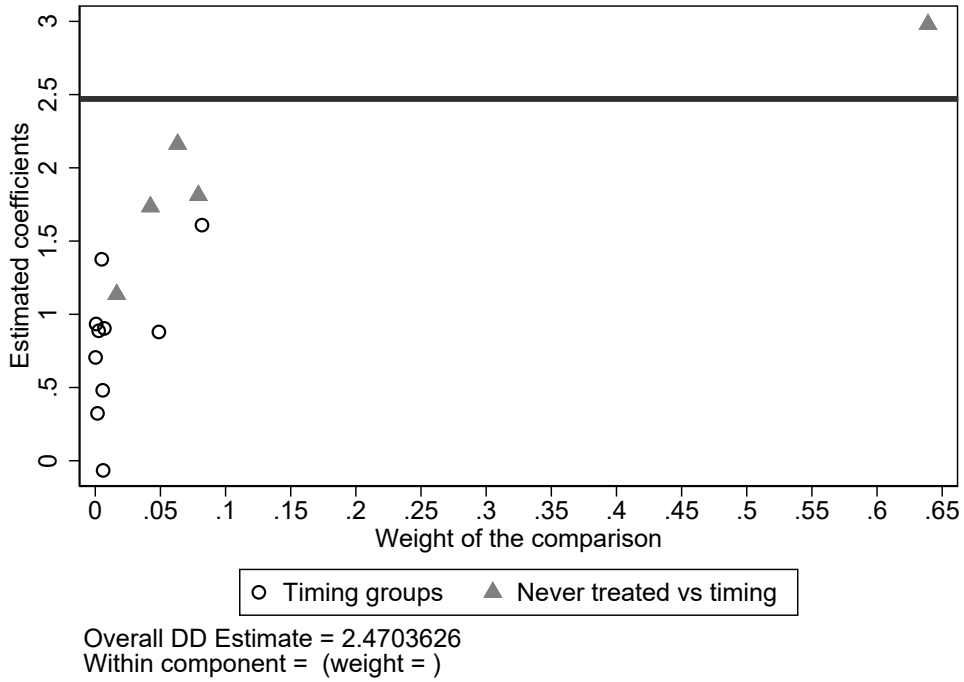


Figure A7: Goodman-Bacon decomposition: weights and estimates. The figure reports the Goodman-Bacon (2021) decomposition of the conventional TWFE estimator based on physicians' formal entry into the IRCCS perimeter. Each marker represents a two-group, two-period difference-in-differences comparison. Circles denote comparisons among cohorts entering the perimeter at different times; triangles denote comparisons between entering cohorts and physicians who never enter the perimeter. The horizontal coordinate reports the weight assigned to each comparison, while the vertical coordinate reports its estimated effect. The horizontal line marks the aggregate TWFE estimate. The decomposition is reported as a diagnostic of the comparisons underlying conventional TWFE; the preferred staggered analysis restricts the sample to eventually treated physicians and reports estimators robust to heterogeneous treatment effects.

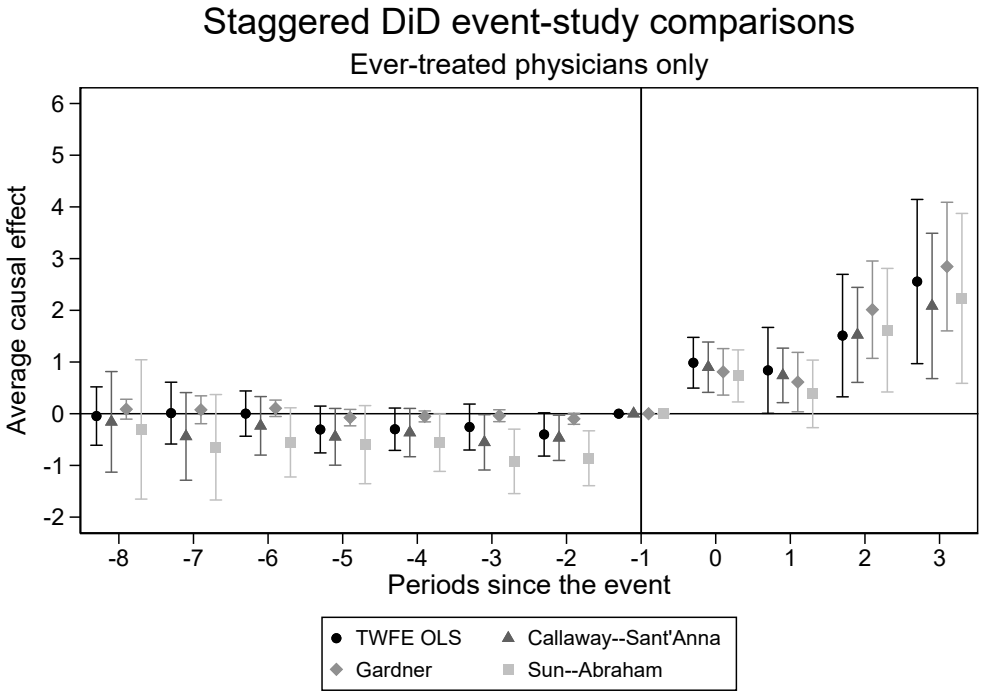
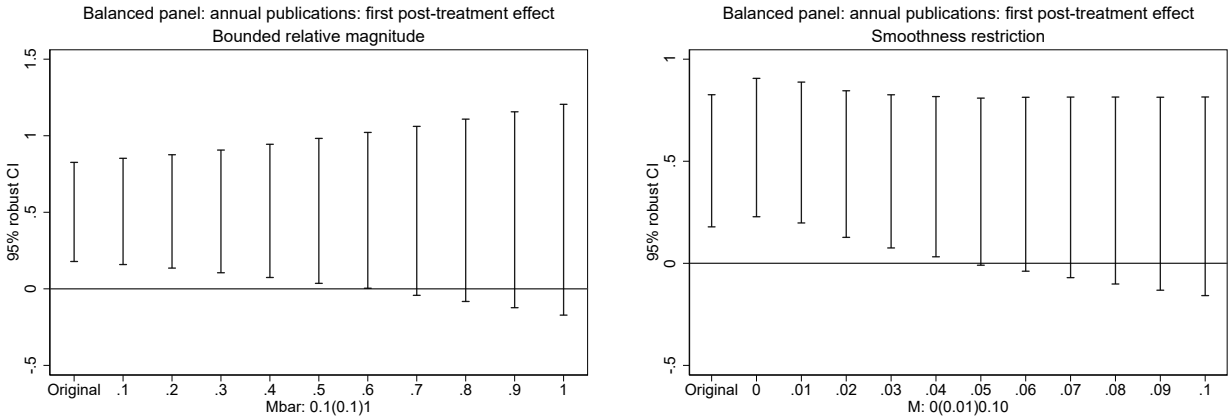
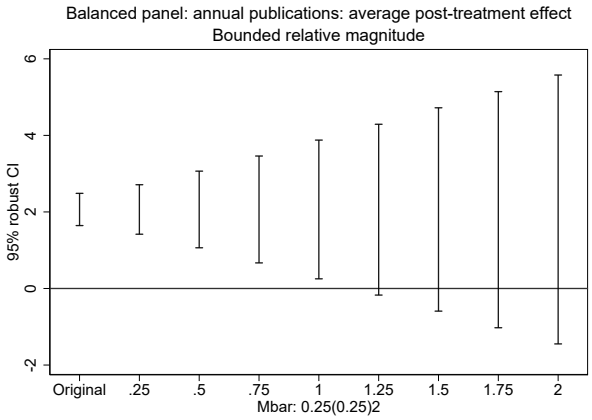


Figure A8: Scientific production around formal entry into the IRCCS perimeter. The figure reports event-study estimates of annual publications relative to physicians' first recorded entry into the IRCCS perimeter, restricting the sample to ever-treated physicians. Conventional TWFE estimates are compared with those of Callaway and Sant'Anna (2021), Gardner (2022), and Sun and Abraham (2021). Event time -1 is the reference period. Vertical bars indicate 95% confidence intervals with physician-clustered standard errors. With no never-treated units, untreated comparisons depend on cohort and calendar time. Sun–Abraham uses the last-treated cohort as the comparison group and therefore excludes 2022, when the last cohort enters the IRCCS perimeter.

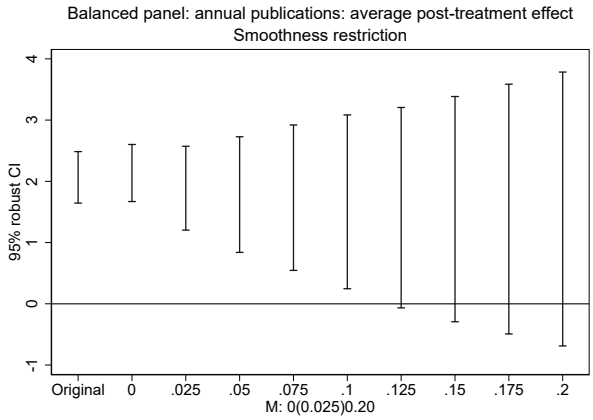


(a) First post-treatment effect: BM

(b) First post-treatment effect: SR



(c) Average post-treatment effect: BM



(d) Average post-treatment effect: SR

Figure A9: Sensitivity to violations of parallel trends. Robust 95% confidence intervals following Rambachan and Roth (2023) for the first post-treatment effect and the equally weighted post-treatment average. BM denotes bounded relative magnitude and SR denotes smoothness; "Original" is the conventional interval.

	Core 2018	Late entrants	Difference	p-value
Mean annual publications before 2018	4.026	1.621	2.405***	0.000
Total publications before 2018	24.155	9.723	14.431***	0.000
No publications before 2018	0.071	0.085	-0.014	0.685
Academic physician in 2017	0.761	0.606	0.155**	0.009
Age in 2017	52.768	50.585	2.183*	0.025

Table A5: Pre-treatment characteristics of initial and late IRCCS entrants. The table compares physicians included in the IRCCS perimeter in 2018 with physicians first recorded as entering after 2018. Publication variables are calculated over 2012–2017. "No publications before 2018" identifies physicians with no publications during the pre-recognition period. Academic status and age are measured in 2017. The difference column reports the mean for the 2018 cohort minus the mean for late entrants. The final column reports the corresponding two-sided p-value. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

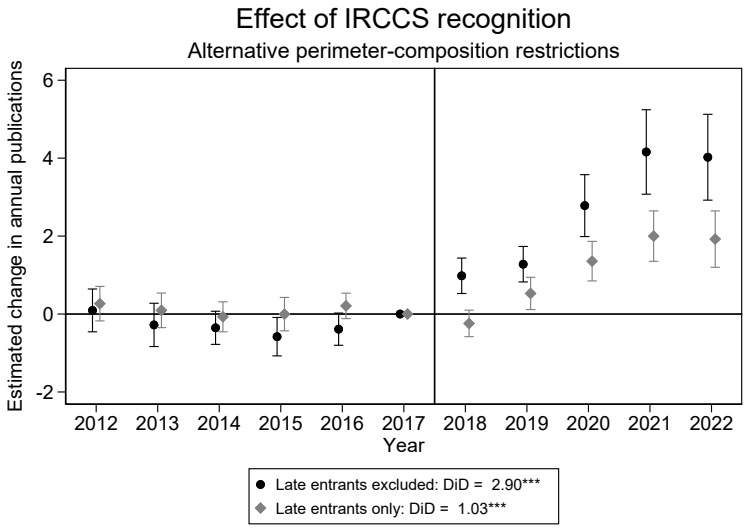


Figure A10: IRCCS recognition under alternative perimeter-composition restrictions. The figure reports event-study estimates under two restrictions on the composition of the IRCCS-treated group. The first specification excludes physicians entering the IRCCS perimeter after 2018 and compares physicians already included at recognition with physicians who never enter the perimeter. The second excludes physicians included in the perimeter in 2018 and compares late entrants with the same never-IRCCS comparison group. IRCCS exposure begins in 2018 and 2017 is the omitted reference year. The corresponding static difference-in-differences coefficients and significance levels are reported in the legend. All specifications include physician and department-by-year fixed effects. Vertical bars report 95% confidence intervals based on standard errors clustered at the physician level.

	(1) (1)	(2) (2)	(3) (3)
Mean annual publications before 2018	-0.035***	-0.036***	-0.033***
	(0.008)	(0.008)	(0.008)
No publications before 2018		-0.076	-0.067
		(0.122)	(0.121)
Academic physician in 2017			-0.079
			(0.085)
Age in 2017			-0.003
			(0.005)
Constant	0.488***	0.497***	0.682***
	(0.041)	(0.044)	(0.229)
Physicians	249	249	249
R-squared	0.085	0.087	0.097

Table A6: Observable correlates of entry into the IRCCS perimeter after 2018. The table reports linear probability models estimated among physicians who eventually enter the IRCCS perimeter. The dependent variable equals one for physicians first recorded as entering after 2018 and zero for physicians already included in 2018. Publication variables are calculated over 2012–2017. Academic status and age are measured in 2017. Heteroskedasticity-robust standard errors are reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, and $^{***}p < 0.01$.

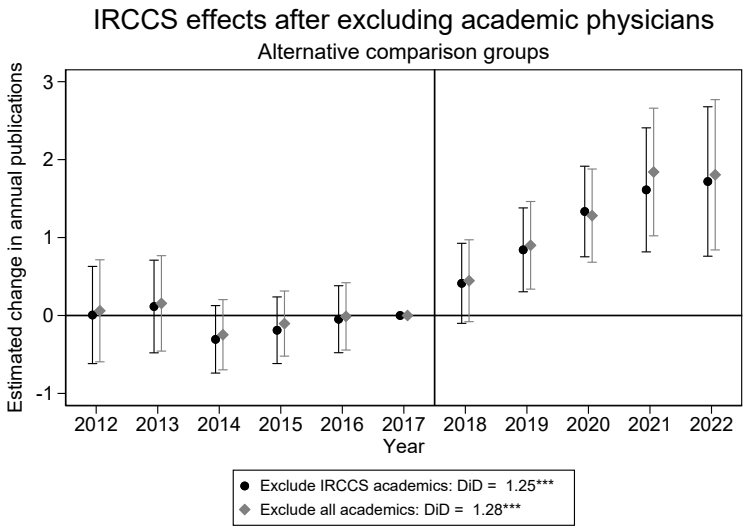


Figure A11: IRCCS recognition under alternative exclusions of academic physicians. Event studies exclude academics from the IRCCS group only (N = 4,315 physician-years) or from both groups (N = 2,880). Models include physician and department-by-year fixed effects and physician-clustered standard errors; 2017 is omitted.

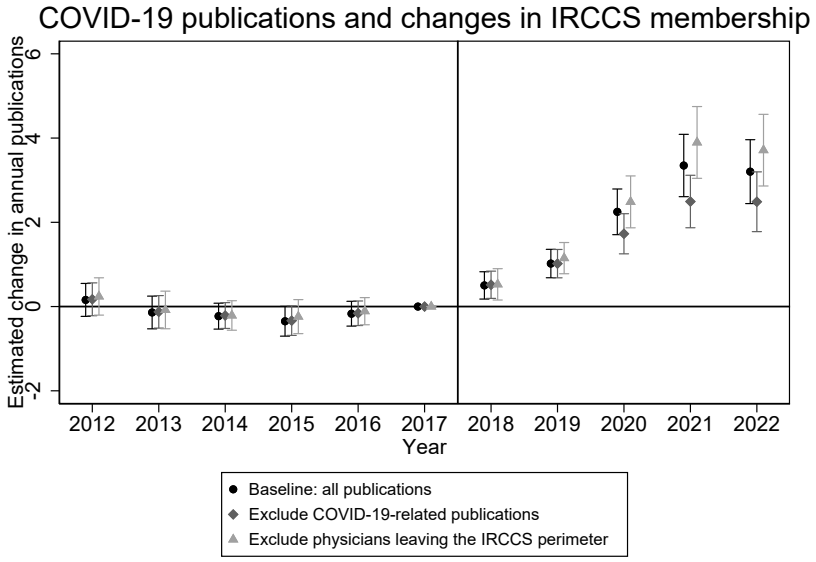


Figure A12: Robustness to COVID-19 publications and changes in IRCCS membership. Event studies report the baseline, publication counts excluding COVID-19 articles, and estimates excluding physicians who leave the IRCCS perimeter. Models use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

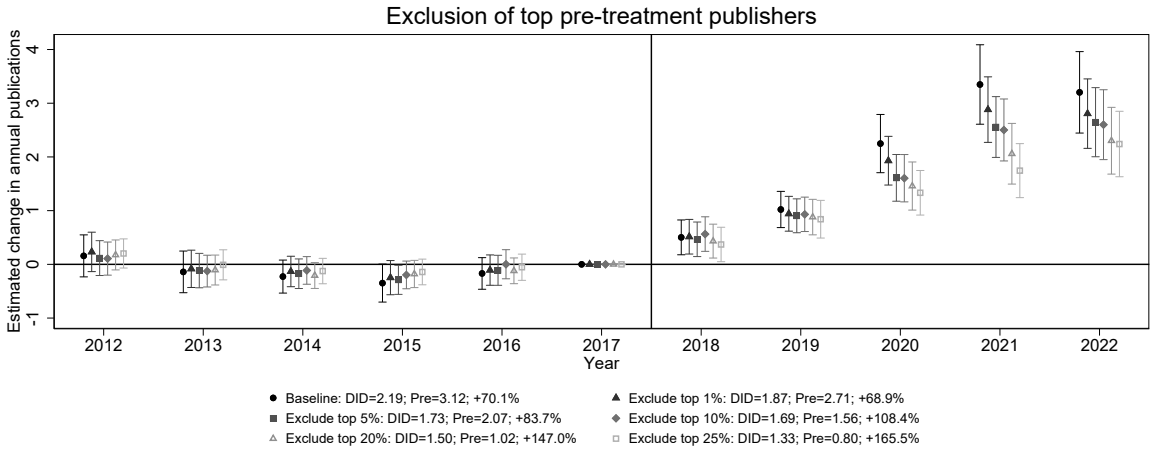


Figure A13: Robustness to the progressive exclusion of highly productive physicians. Event studies symmetrically trim the pooled physician-level distribution of average annual publications in 2012–2017 and remove each excluded physician's full history. Models use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

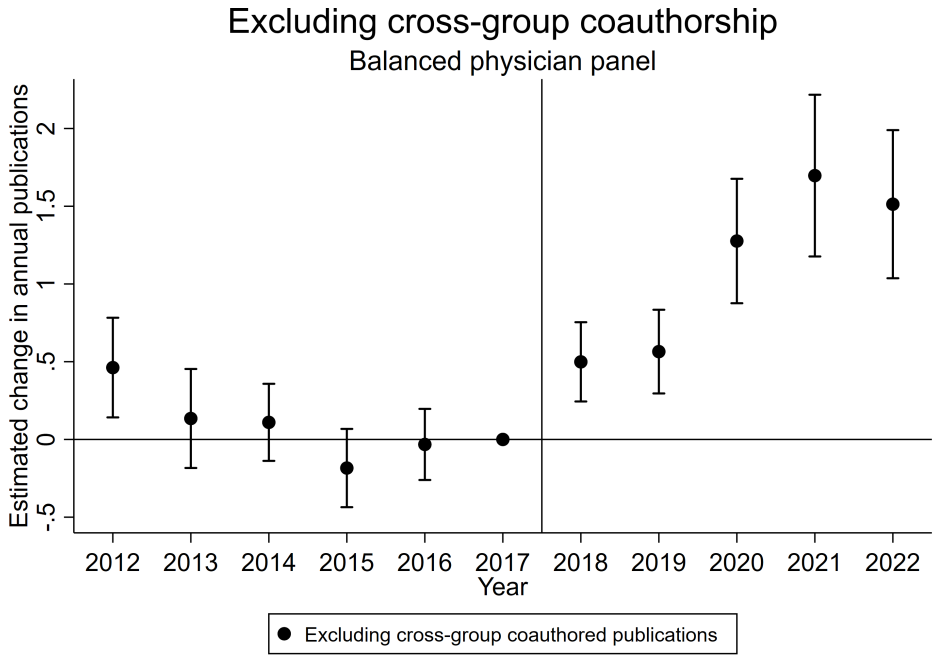


Figure A14: Publication effects after excluding cross-group coauthorship. Event-study outcome excludes papers jointly authored by ever-IRCCS and internal never-IRCCS physicians. Models use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

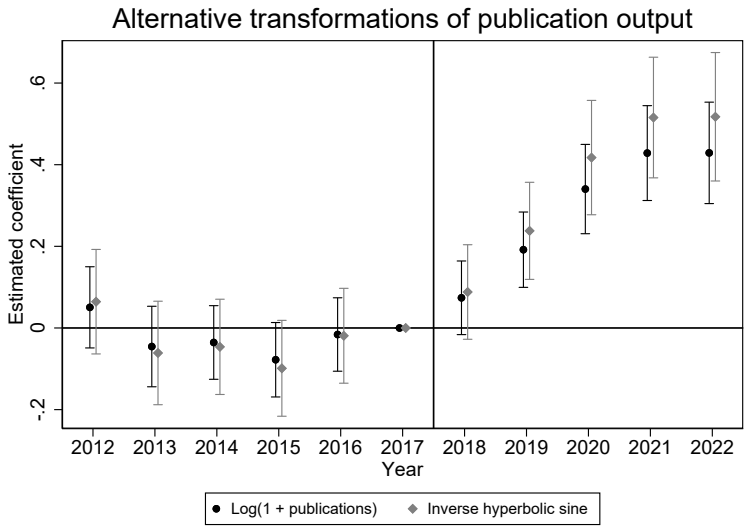


Figure A15: Alternative transformations of publication output. Event studies use $\log(1 + \text{publications})$ and the inverse hyperbolic sine of annual publications. Models use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is omitted.

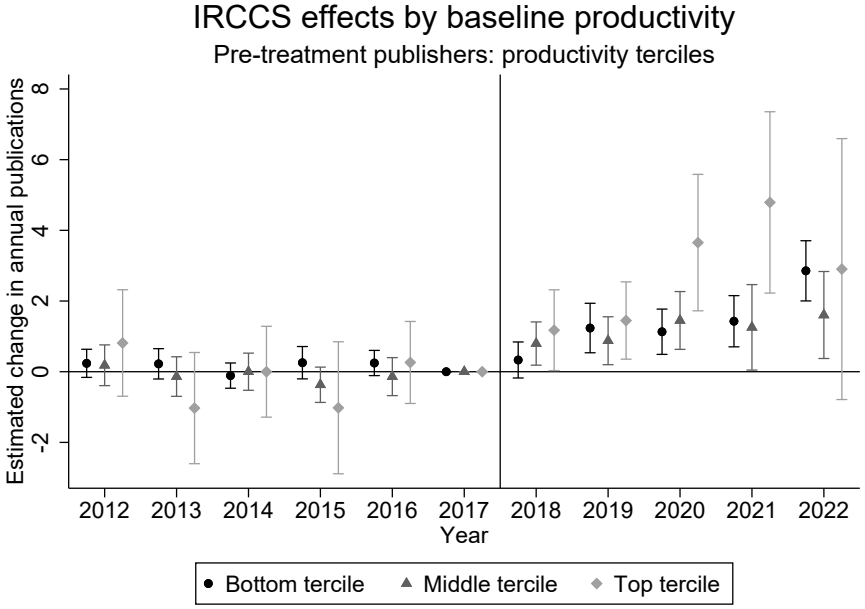


Figure A16: Effect of IRCCS recognition by baseline productivity among pre-treatment publishers. The sample includes physicians publishing in 2012–2017; terciles are based on their average annual output in that period. Event studies use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is omitted.

Macro-area	WoS categories included
Biology	Biochemistry; Biotechnology; Cell Biology; Genetics; Life Sciences Biomedicine; Microbiology; Biophysics; Developmental Biology; Evolutionary Biology; Physiology; Agriculture; Fisheries; Veterinary Sciences; Zoology
Cardiology/Cardiovascular Diseases	Cardiology
Chemistry/Physics	Chemistry; Electrochemistry; Crystallography; Materials Science; Physics; Polymer Science; Spectroscopy; Acoustics; Optics
Diagnostics/Imaging	Radiology; Pathology; Medical Laboratory Technology
Infectious Diseases/Immuno-Allergology	Infectious Diseases; Virology; Parasitology; Tropical Medicine; Mycology; Immunology; Allergy
Internal Medicine (Generic)	General and Internal Medicine; Emergency Medicine; Integrative and Complementary Medicine; Research and Experimental Medicine
Internal Medicine (Specialties)	Gastroenterology; Endocrinology; Respiratory System; Rheumatology; Geriatrics; Dermatology; Nutrition and Dietetics
Mother/Child	Obstetrics; Pediatrics; Reproductive Biology
Neuroscience/Psychiatry/Psychology	Neurosciences; Psychiatry; Psychology
Oncology/Hematology	Oncology; Hematology
Pharmacology	Pharmacology; Toxicology; Substance Abuse
Public Health/Health Systems	Health Care Sciences and Services; Public, Environmental and Occupational Health; Business and Economics; Public Administration; Biomedical Social Sciences; Environmental Sciences
Rehabilitation/Care	Rehabilitation; Nursing; Sport Sciences; Audiology
Social Sciences and Humanities	Behavioral Sciences; Legal Medicine; Medical Ethics; Sociology; Social Sciences - Other Topics; Family Studies; Educational Research; Religion; Criminology; Anthropology; History and Philosophy of Science
Surgery/Procedural	Surgery; Anesthesiology; Orthopedics; Otorhinolaryngology; Urology; Transplantation; Ophthalmology; Dentistry
Technology/Engineering	Medical Laboratory Technology; Instruments and Instrumentation; Engineering; Computer Science; Mechanics; Metallurgy and Metallurgical Engineering; Microscopy; Nuclear Science and Technology; Robotics; Science and Technology - Other Topics; Food Science

Table A7: Classification of research areas into macro-categories based on Web of Science (WoS) subject categories.

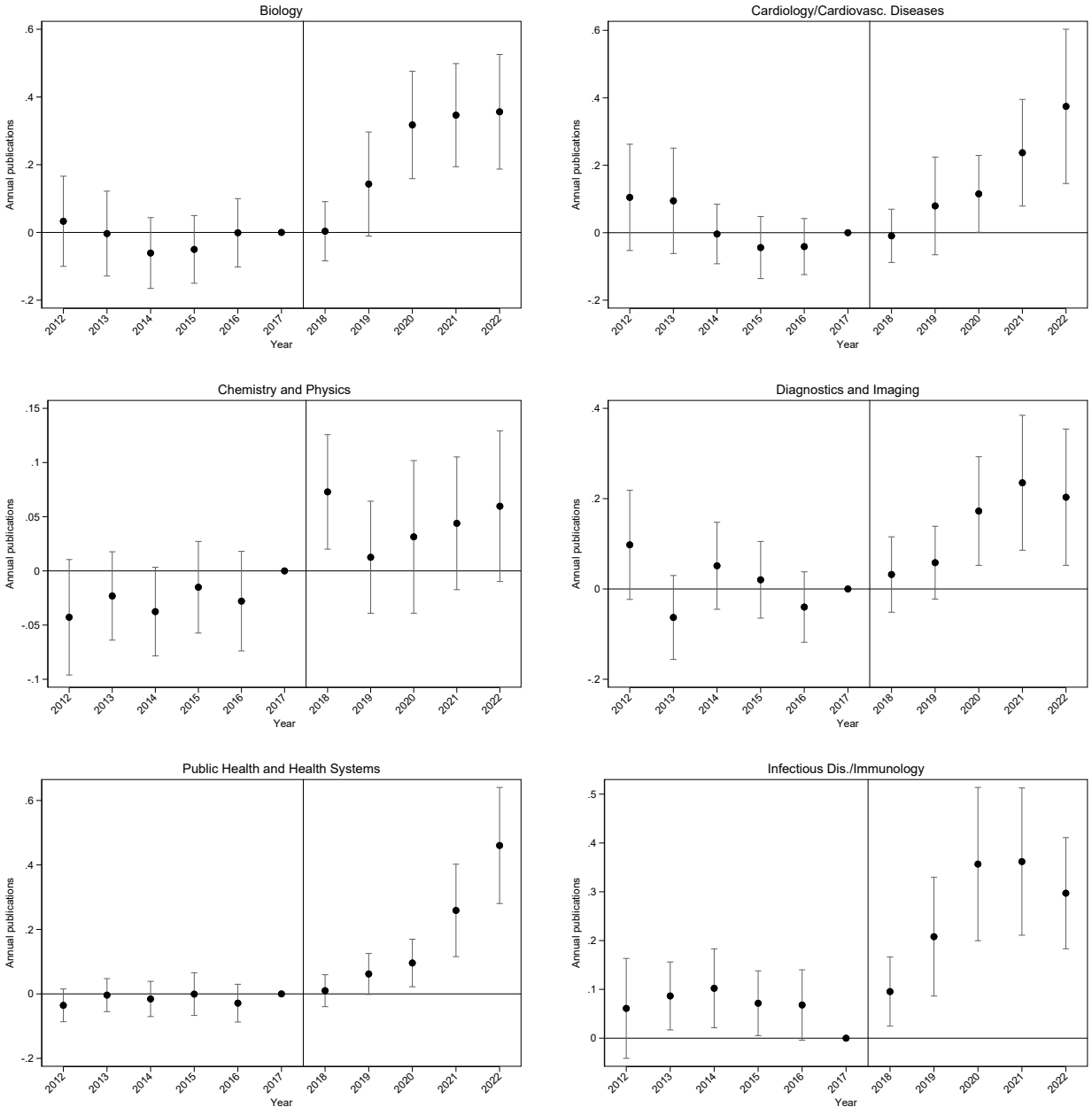


Figure A17: Effect of IRCCS recognition on field-specific publication output: first group of scientific fields

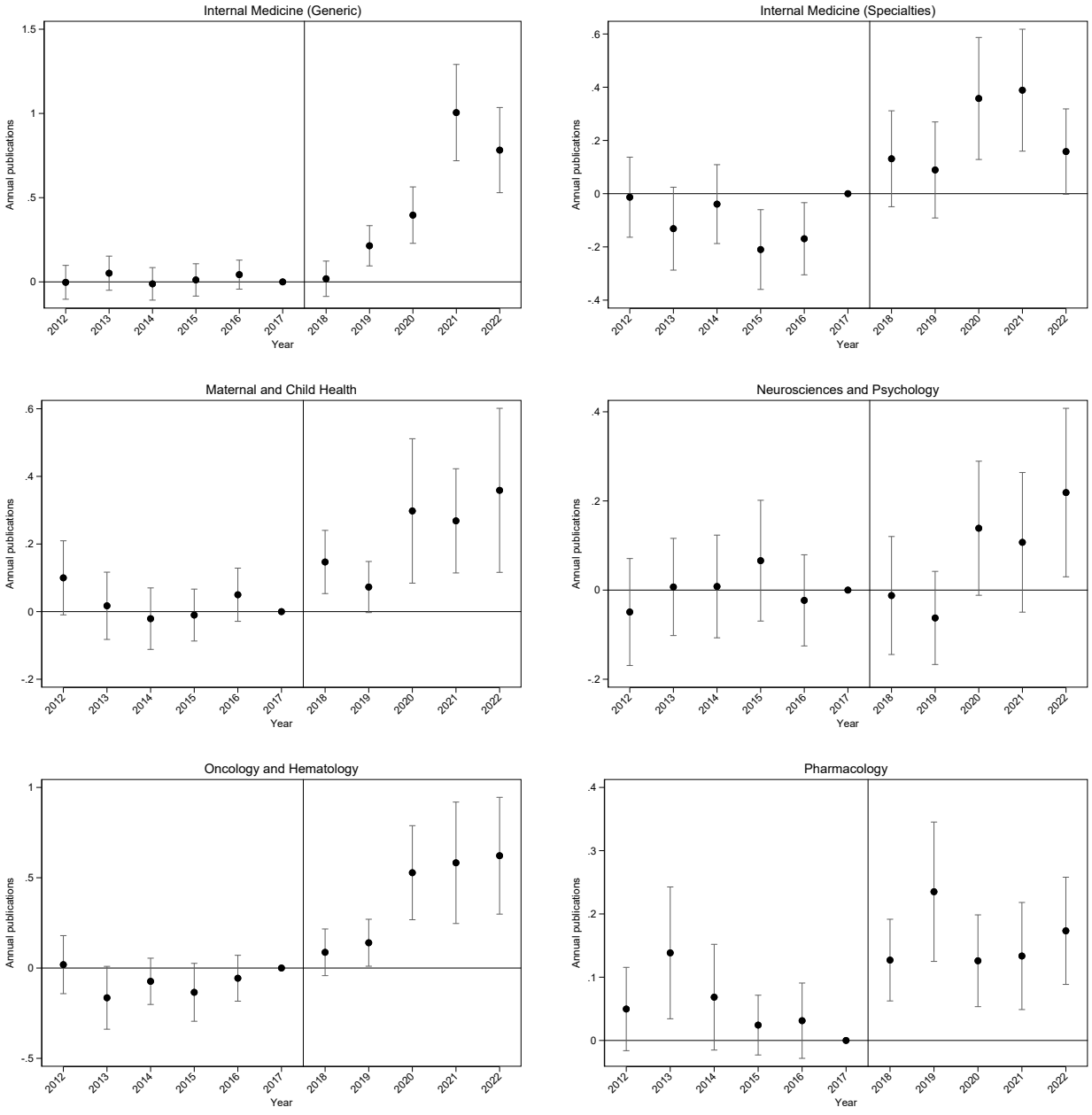


Figure A18: Effect of IRCCS recognition on field-specific publication output: second group of scientific fields

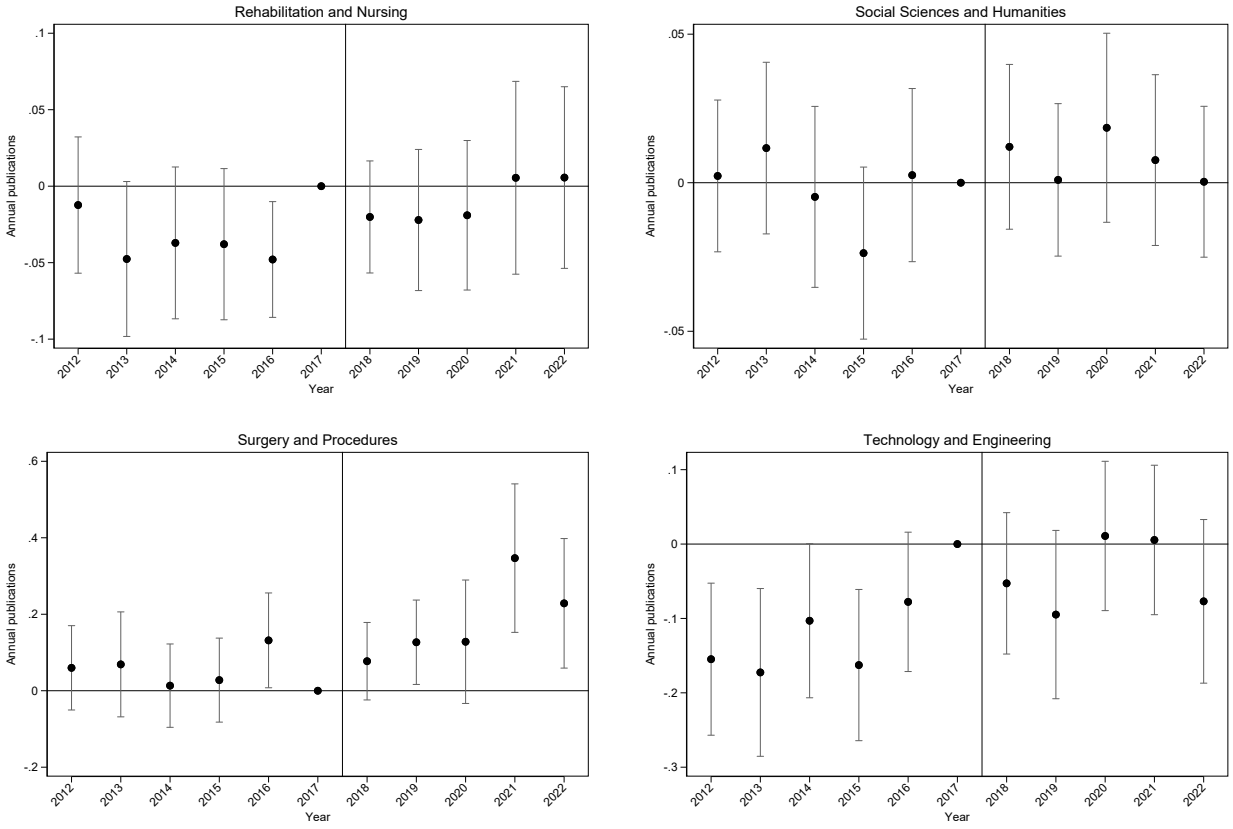


Figure A19: Effect of IRCCS recognition on field-specific publication output: third group of scientific fields. Figures A17– A19 report annual publications by Web of Science macro-field; papers may enter multiple fields. Models use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

Table A8: Effect of IRCCS recognition on physicians' research-portfolio shares. Rows are separate difference-in-differences models for annual publication shares by field. Columns report the estimate, clustered standard error, conventional and Romano–Wolf adjusted p-values, pre-treatment test, treated pre-mean, and observations. Models include physician and department-by-year fixed effects.

Field	Coef.	SE	p-value	RW p-value	Pre-test	Treated pre-mean	Obs.
Biology	-0.008	(0.014)	0.557	0.963	0.458	0.075	3,325
Cardiology/Cardiovasc. Diseases	-0.021	(0.021)	0.321	0.863	0.262	0.069	3,325
Chemistry and Physics	-0.004	(0.007)	0.539	0.963	0.509	0.009	3,325
Diagnostics and Imaging	-0.021	(0.016)	0.180	0.572	0.364	0.071	3,325
Infectious Dis./Immunology	0.002	(0.012)	0.887	0.977	0.667	0.039	3,325
Internal Medicine (Generic)	0.011	(0.028)	0.707	0.970	0.763	0.075	3,325
Internal Medicine (Specialties)	-0.018	(0.025)	0.467	0.944	0.086	0.157	3,325
Maternal and Child Health	0.007	(0.016)	0.673	0.970	0.944	0.053	3,325
Neurosciences and Psychology	-0.032*	(0.017)	0.058	0.236	0.028	0.120	3,325
Oncology and Hematology	-0.015	(0.019)	0.438	0.944	0.893	0.112	3,325
Pharmacology	0.027	(0.017)	0.106	0.355	0.154	0.027	3,325
Public Health and Health Systems	0.032*	(0.017)	0.069	0.280	0.207	0.017	3,325
Rehabilitation and Nursing	0.012	(0.014)	0.397	0.926	0.141	0.010	3,325
Social Sciences and Humanities	0.000	(0.004)	0.973	0.977	0.177	0.006	3,325
Surgery and Procedures	0.059*	(0.034)	0.086	0.300	0.250	0.124	3,325
Technology and Engineering	-0.029	(0.020)	0.144	0.473	0.257	0.037	3,325

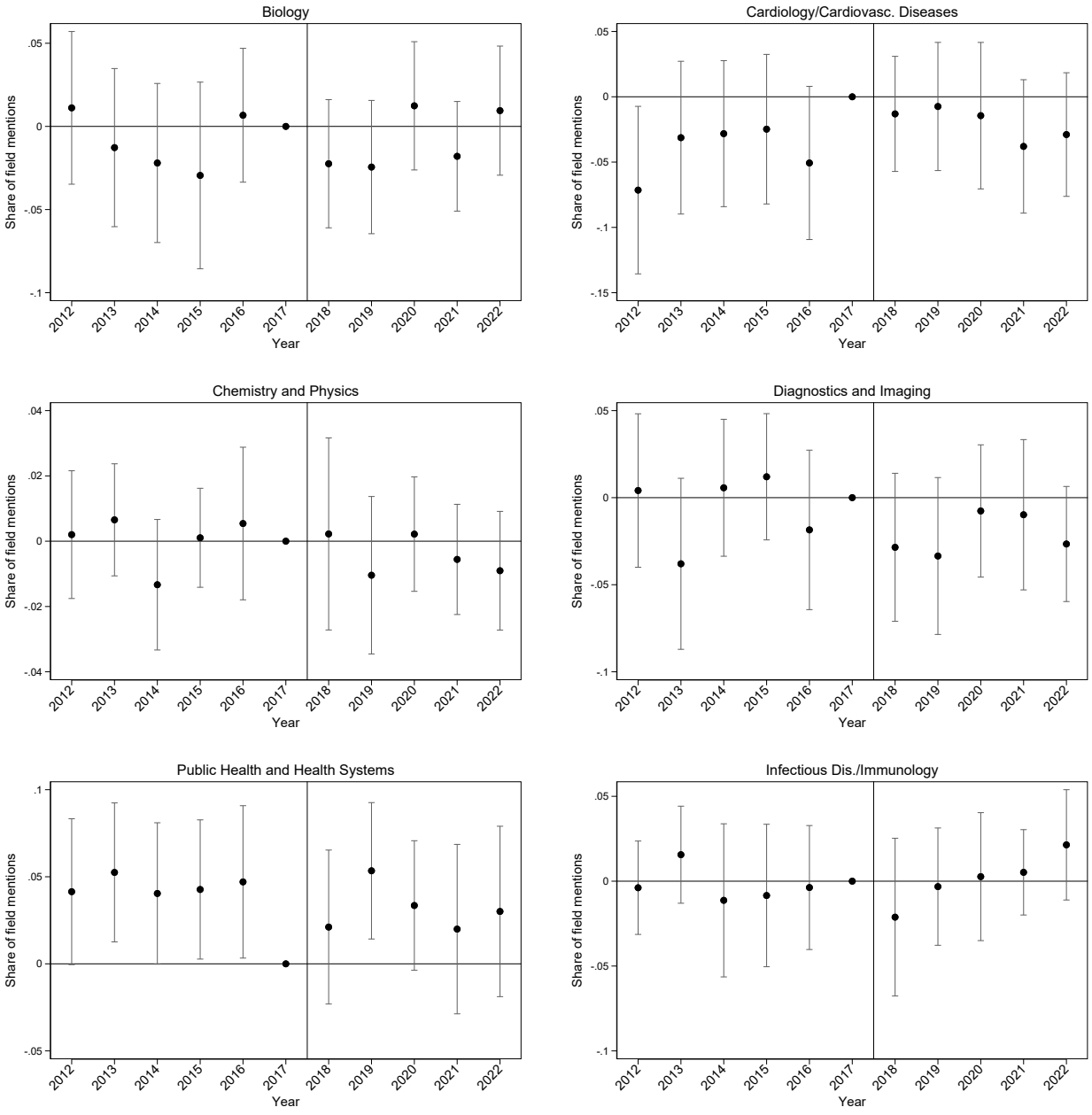


Figure A20: Effect of IRCCS recognition on physicians' research-portfolio shares: first group of scientific fields

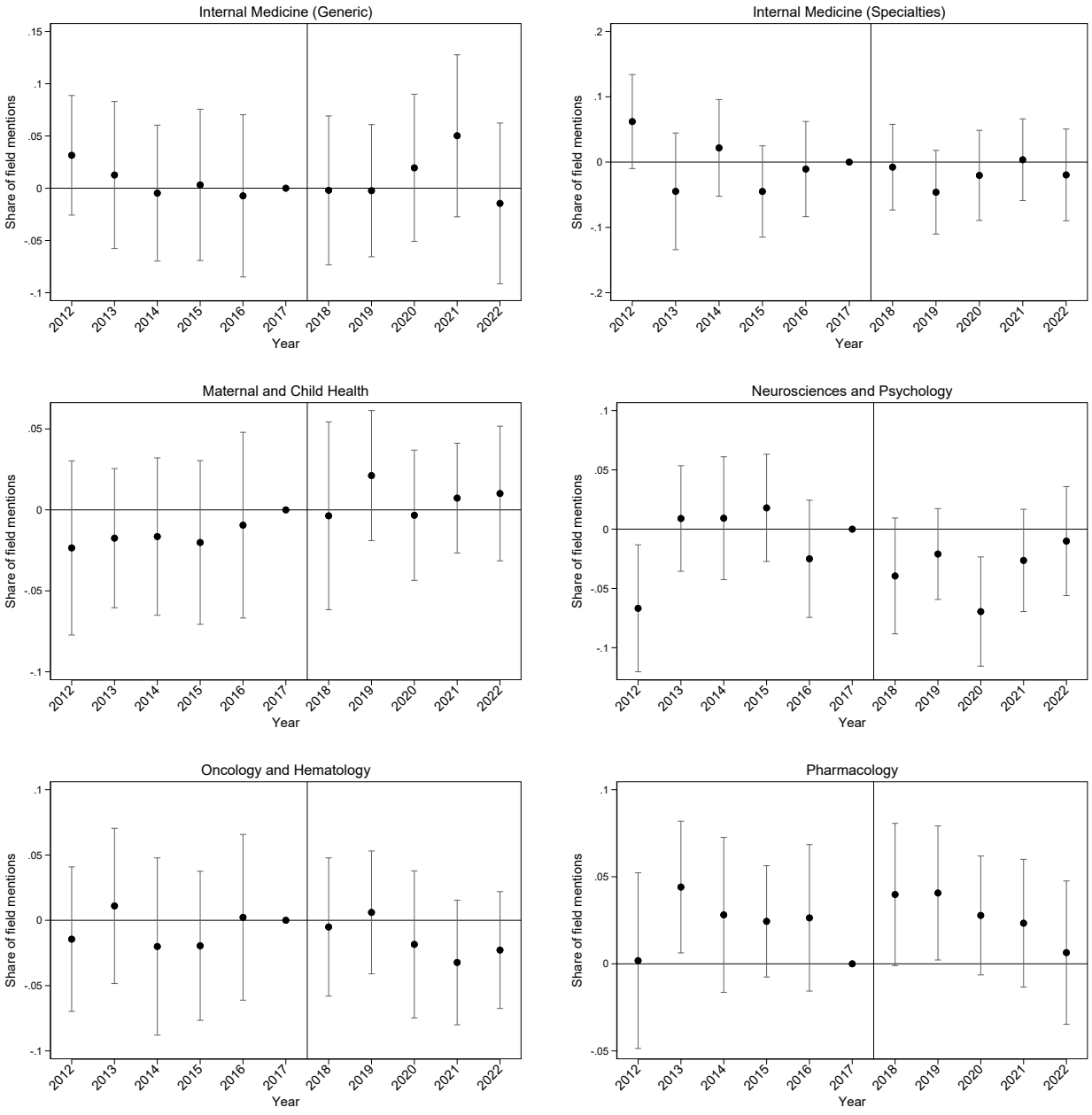


Figure A21: Effect of IRCCS recognition on physicians' research-portfolio shares: second group of scientific fields

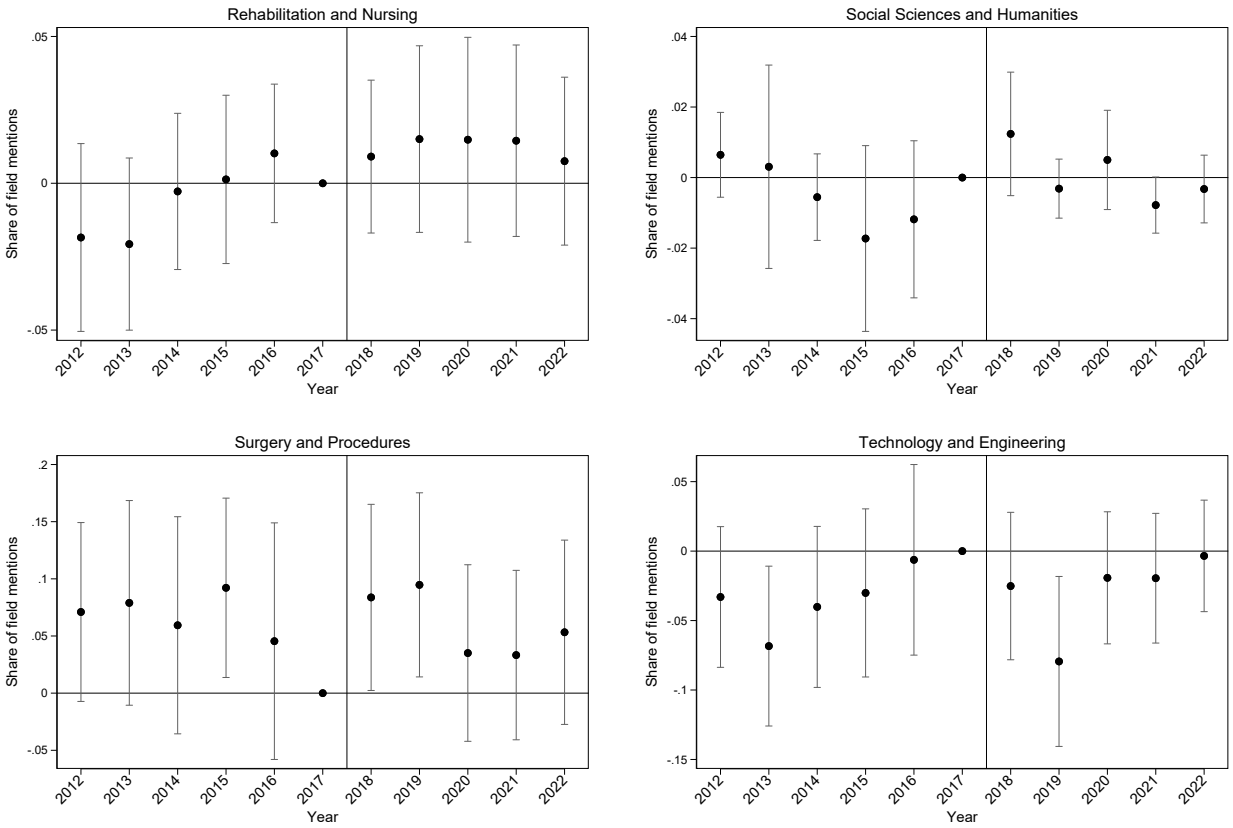
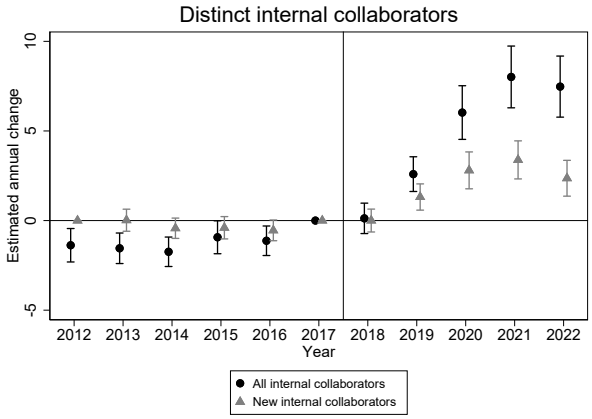
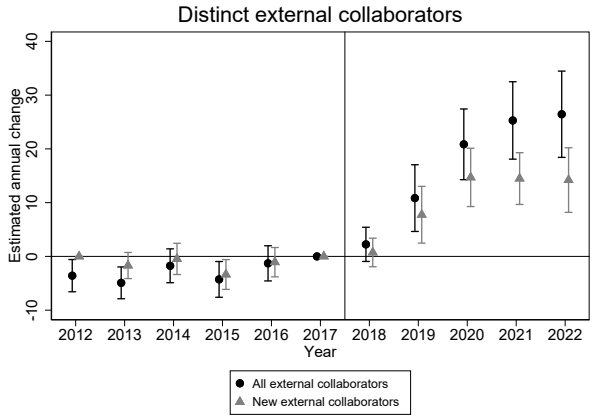


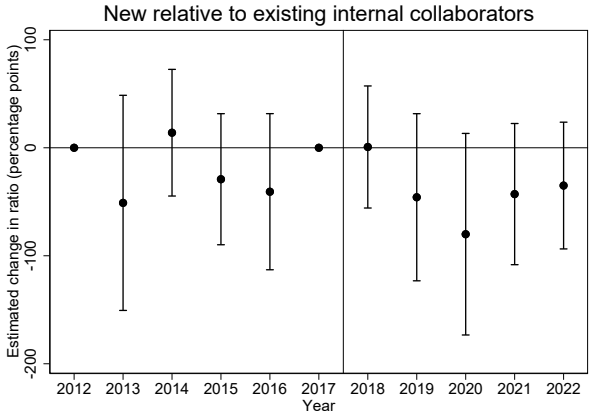
Figure A22: Effect of IRCCS recognition on physicians' research-portfolio shares: third group of scientific fields. Figures A20– A22 use annual publication shares by Web of Science macro-field and exclude physician-years without publications; papers may enter multiple fields. Models use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.



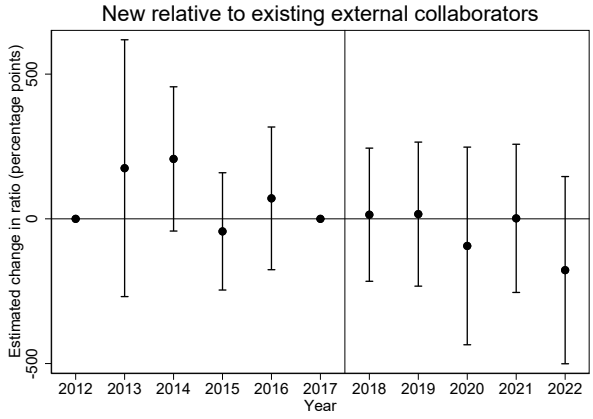
(a) Distinct internal collaborators



(b) Distinct external collaborators



(c) New relative to existing internal collaborators



(d) New relative to existing external collaborators

Figure A23: IRCCS recognition and distinct collaborators. Panels (a)–(b) report total and new internal and external collaborators; Panels (c)–(d) report ratios of new to previously observed collaborators. Internal collaborators are physicians in the hospital roster. Models use the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is normalized to zero.

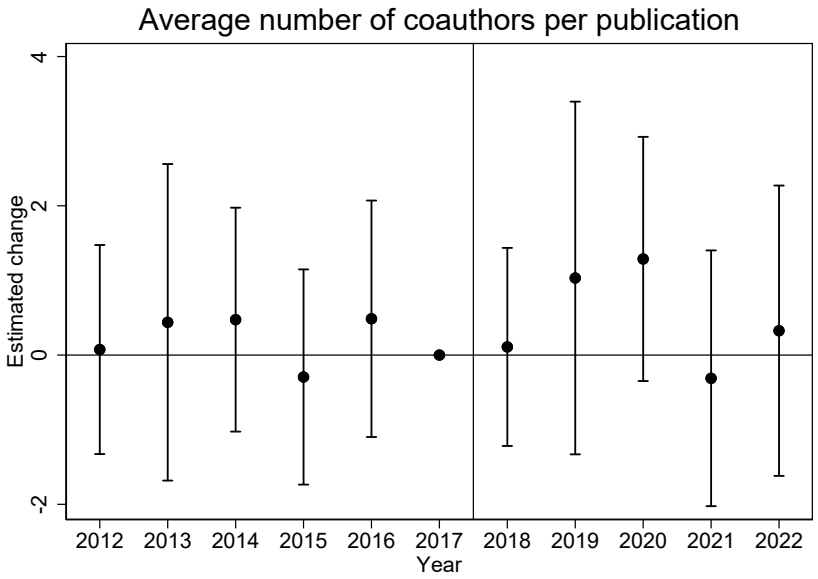
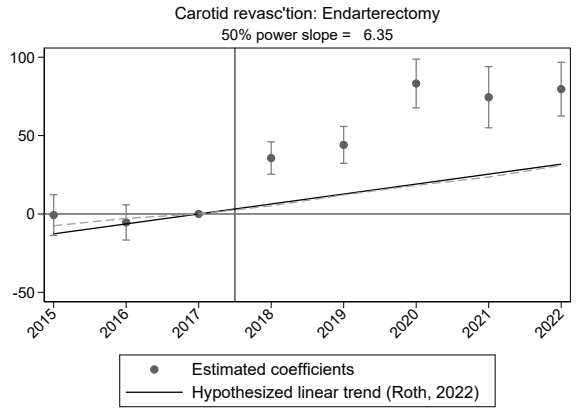
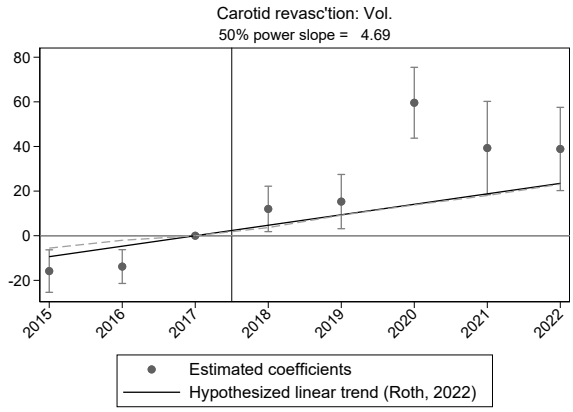


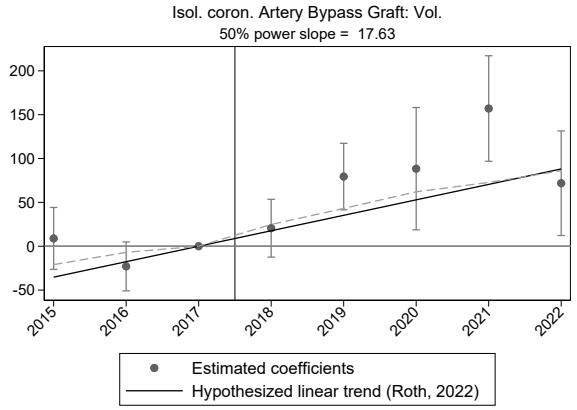
Figure A24: Average number of coauthors per publication. The outcome is the physician-year mean and excludes years without publications. The model uses the balanced panel, physician and department-by-year fixed effects, and physician-clustered standard errors; 2017 is omitted.



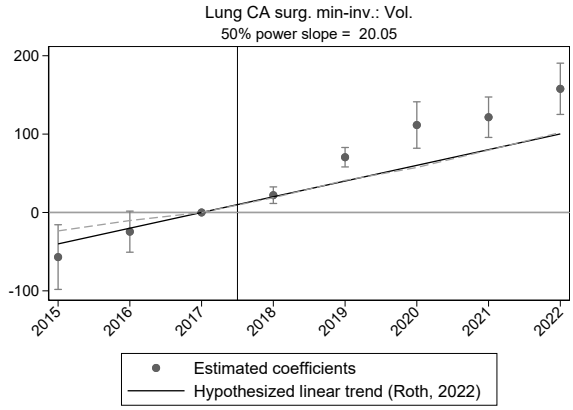
(a) Carotid endarterectomy



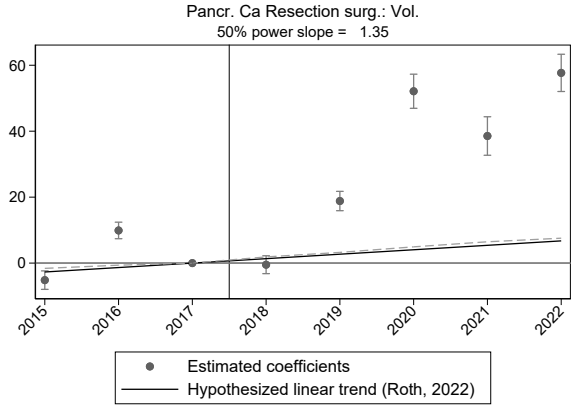
(b) Carotid revascularization volume



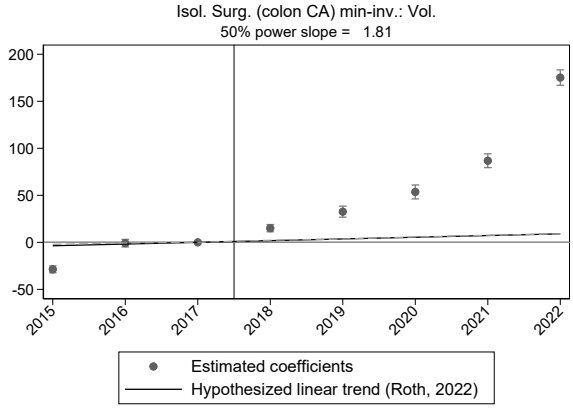
(c) Coronary artery bypass grafting



(d) Minimally invasive lung cancer surgery

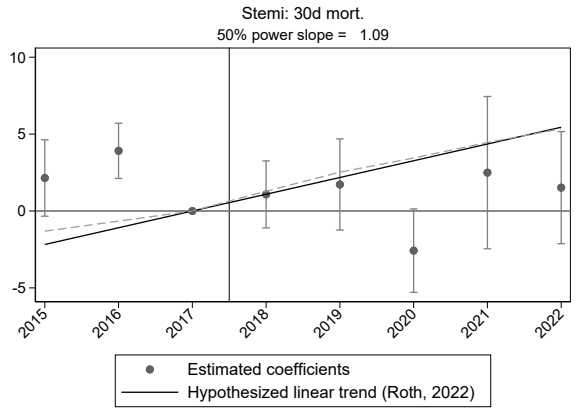


(e) Pancreatic cancer resection

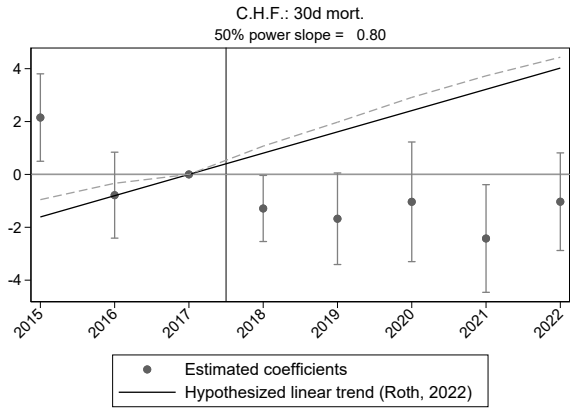


(f) Minimally invasive colon cancer surgery

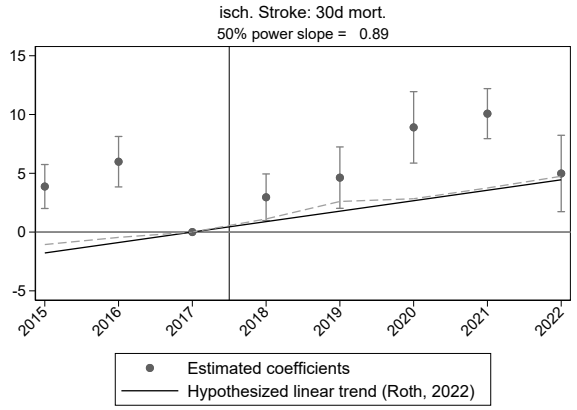
Figure A25: Selected high-complexity and technology-intensive procedure volumes. Hospital-level event studies compare the treated hospital with non-IRCCS hospitals in Lazio and include hospital and year fixed effects. The reference year is 2017 and confidence intervals use hospital-clustered standard errors. Solid and dashed lines show the detectable linear trend and conditional expected coefficients from Roth (2022).



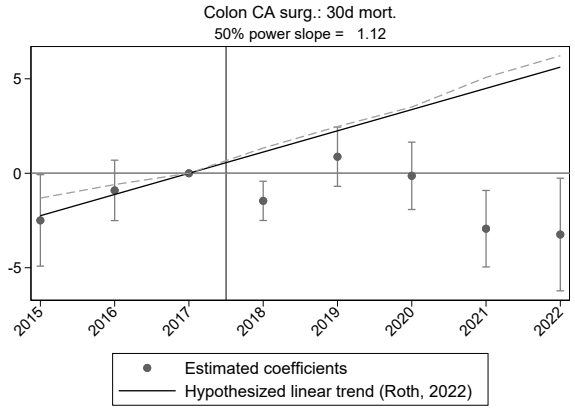
(a) STEMI: 30-day mortality



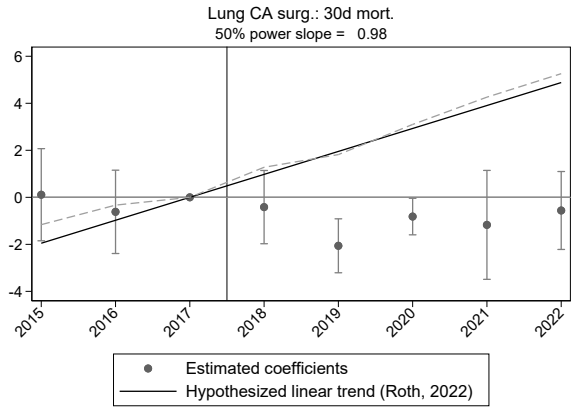
(b) Heart failure: 30-day mortality



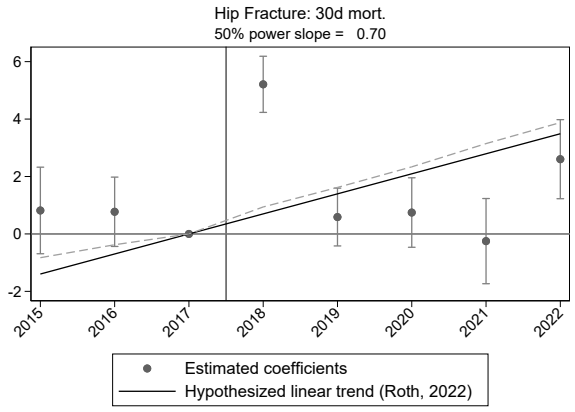
(c) Ischemic stroke: 30-day mortality



(d) Colon cancer surgery: 30-day mortality



(e) Lung cancer surgery: 30-day mortality



(f) Hip fracture: 30-day mortality

Figure A26: Selected hospital-quality indicators. Hospital-level event studies compare the treated hospital with non-IRCCS hospitals in Lazio and include hospital and year fixed effects. The reference year is 2017 and confidence intervals use hospital-clustered standard errors. Solid and dashed lines show the detectable linear trend and conditional expected coefficients from Roth (2022).

Outcome	Estimate	SE	p	Pre-trend p	Roth slope	Treated pre-mean
Isolated coronary artery bypass grafting: procedure volume	88.099	(22.559)	0.003	0.000	17.343	291.00
ST-elevation myocardial infarction: procedure volume	-8.113	(3.001)	0.009	0.000	3.809	245.67
Ischemic stroke: procedure volume	0.028	(3.154)	0.993	0.000	3.265	503.67
Congestive heart failure: procedure volume	-21.095	(12.035)	0.084	0.001	11.060	514.67
Pancreatic cancer resection surgery: procedure volume	31.826	(1.687)	0.000	0.000	1.348	87.67
Isolated colon cancer surgery: procedure volume	79.406	(2.326)	0.000	0.000	1.628	361.33
Lung cancer surgery: procedure volume	39.045	(14.525)	0.014	0.345	16.252	252.00

Table A9: Selected PNE cardiovascular and oncological activity volumes. Rows report separate hospital-level difference-in-differences estimates for the treated hospital relative to non-IRCCS hospitals in Lazio. Models include hospital and year fixed effects, with standard errors clustered by hospital. Pre-trend p-values jointly test the 2015–2016 coefficients relative to 2017; Roth slopes report the linear differential trend against which this test has 50% power.

Outcome	Estimate	SE	p	Pre-trend p	Roth slope	Treated pre-mean
ST-elevation myocardial infarction: 30-day mortality	-1.171	(0.915)	0.217	0.002	1.089	9.69
Ischemic stroke: 30-day mortality	3.024	(0.727)	0.001	0.000	0.889	10.28
Congestive heart failure: 30-day mortality	-1.947	(0.567)	0.001	0.000	0.804	8.67
Pancreatic cancer resection surgery: 90-day mortality	-11.597	(1.127)	0.009	0.970	9.691	5.46
Colon cancer surgery: 30-day mortality	-0.246	(0.570)	0.669	0.141	1.122	1.79
Lung cancer surgery: 30-day mortality	-0.836	(0.715)	0.295	0.292	0.976	0.70

Table A10: Selected PNE short-term mortality indicators. Rows report separate hospital-level difference-in-differences estimates for the treated hospital relative to non-IRCCS hospitals in Lazio. Models include hospital and year fixed effects, with standard errors clustered by hospital. Pre-trend p-values jointly test the 2015–2016 coefficients relative to 2017; Roth slopes report the linear differential trend against which this test has 50% power.